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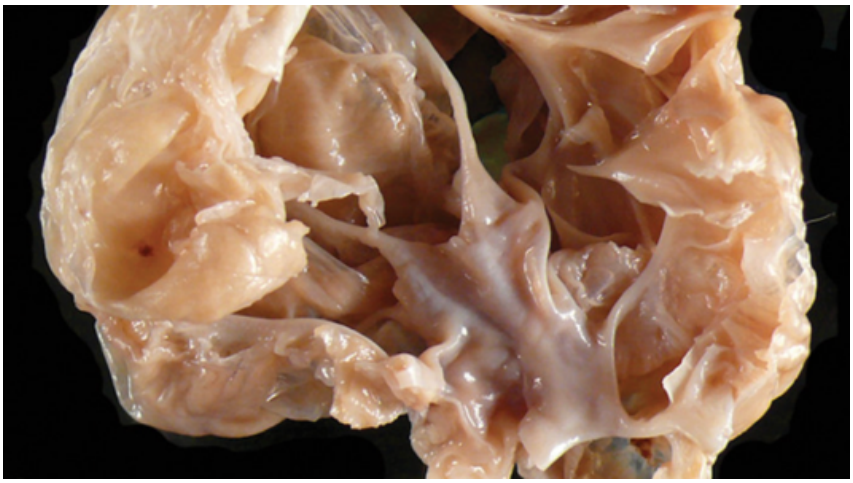
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HY Renal – by Dr Mike Mehlman

Basic HY Renal lab values you must know for USMLE	
Blood urea nitrogen (BUN)	- Should be <20 mg/dL. - High urea nitrogen in the blood is called azotemia. - "Uremia" technically means "urea in the blood," but is used to refer to patients who have symptomatic renal failure with poor lab values. This has absolutely zero to do with hyperuricemia (gout) and is sometimes confused by students.
Creatinine (Cr)	- 0.7-1.2 mg/dL - Once you've hit a creatinine of 2, you've lost ~90% of your renal function.
Potassium (K^+)	- 3.5-5.0 mEq/L. - Will always be elevated in renal failure on USMLE. - This is because the kidney normally secretes (i.e., excretes through tubular walls into the urine) K^+ in the cortical collecting duct under the action of aldosterone. So if the kidney is fucked up, we can't do that, so serum K^+ rises.
Sodium (Na^+)	- 135-145 mEq/L. - Can be variable in renal failure as per my observation on NBMEs. - Aldosterone normally reabsorbs sodium in the cortical collecting duct in exchange for potassium being secreted, so one might think it should automatically be low if the kidney can't reabsorb it. But the kidney has a multitude of sodium regulation mechanisms (i.e., vasopressin), so Na^+ can be normal.
Calcium (Ca^{2+})	- 8.4-10.2 mg/dL. - Always ↓ in renal failure. This is due to 2 reasons: 1) Ca^{2+} is normally reabsorbed in the late-distal convoluted tubule (DCT), so if we can't do that, serum Ca^{2+} falls. 2) Proximal convoluted tubule (PCT) of the kidney is normally where 1α-hydroxylase, under the action of PTH, activates inactive 25-OH-D3 into active 1,25(OH)2-D3. The latter then goes to the small bowel, where it ↑ absorption of Ca^{2+}. So if we can't do that, serum Ca^{2+} falls. - Low serum Ca^{2+} means ↓ negative feedback at the Ca^{2+}-sensing receptors at the parathyroid glands → PTH goes up. We call this secondary hyperparathyroidism, which means PTH is high due to a cause external to the parathyroid glands themselves. You need to know secondary hyperparathyroidism = renal failure.
Phosphate (PO_4^{3-})	- 3.0-4.5 mEq/L. - Always ↑ in renal failure. - This is for 2 reasons: 1) The kidney cannot filter it out sufficiently. Lots of phosphate is normally filtered by healthy kidneys. 2) PTH normally acts to ↓ phosphate reabsorption in the PCT via downregulation of 3 different types of phosphate pumps. Since the function of PTH is impaired, the kidney *reabsorbs too much* phosphate, so serum levels rise. - Combo of ↓ Ca^{2+} and ↑ PO_4^{3-} in renal failure = secondary hyperparathyroidism.
Bicarbonate (HCO_3^-)	- 22-28 mEq/L. - Always ↓ in renal failure. - This is for 2 reasons: 1) The kidney cannot reabsorb bicarb in the PCT as well, so ↑ bicarb in the urine → ↓ bicarb in serum → metabolic acidosis. 2) The kidney cannot secrete protons (H^+) in the cortical collecting duct as readily → retention of H^+ mops up HCO_3^- → ↓ serum bicarb. - Renal failure = Uremia = the U in MUDPILES for high anion-gap metabolic acidoses. (If you need help on the acid-base stuff, do my HY Arrows PDF).

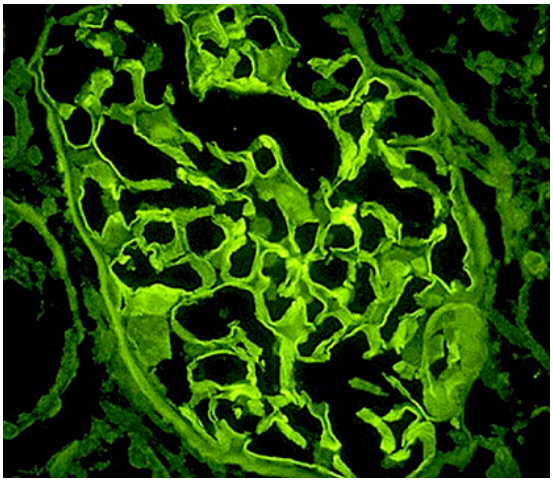
Afferent vs efferent regulation	
Afferent arterioles	- Go to the kidney. - Patency maintained by **vasodilating prostaglandins.** - NSAIDs inhibit prostaglandin synthesis → ↓ dilation of afferent arterioles → ↓ renal blood flow. - NSAIDs don't actively constrict the afferent arterioles. This is incorrect wording. They just merely ↓ vasodilation / ↓ diameter of the afferent arterioles.
Efferent arterioles	- Efferent (Exit) leave the kidney. - Angiotensin II normally constricts these. - ACEi/ARBs therefore ↑ diameter of the efferent arterioles. They are not dilating them; they are merely ↓ constriction of them. As mentioned above, prostaglandins will actively dilate the afferent arterioles. - Some students might say, "Well doesn't AT II act on the afferent arterioles as well?" In theory, a very minor amount, but on USMLE, this will get you questions wrong. You want to think:
Summary	AT II = constricts efferent; ACEi/ARBs = ↓ constriction of efferent (↑ diameter). Prostaglandins = dilate afferent; NSAIDs = ↓ dilation of afferent (↓ diameter).

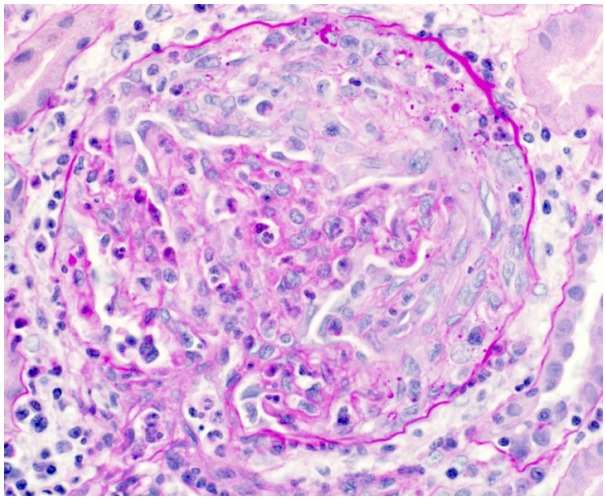
Pre-, intra-, and post-renal azotemias	
BUN/Cr, FENa	- BUN/Cr = Blood urea nitrogen to creatinine ratio. - FENa = fractional excretion of sodium (i.e., amount of sodium in the urine). - There's no "normal" range you should be looking for here for either of these. - What matters is that when there is an actual kidney problem, we look at the values to help us figure out what the diagnosis is. But we never actually say, "Oh the BUN/Cr or FENa is out of the normal range." - BUN/Cr >20 and FENa <1% mean pre-renal azotemia. - BUN/Cr <20 and FENa >1% just simply mean "not pre-renal." - The notion of intra- vs post-renal having slightly different ranges is nonsense for USMLE. - Also, what I've come to learn from NBME Qs is that only ~9/10 times the BUN/Cr will be what we expect. There are a couple 2CK NBME Qs that give BUN/Cr <20 for pre-renal azotemia and >20 for acute tubular necrosis. If the NBME ever "breaks the rules" this way, the vignette will be overwhelmingly obvious what the diagnosis is anyway.
Pre-renal	- Pre-renal azotemia means there is renal pathology because of reduced renal blood flow over the sub-acute to chronic time frame. The kidney will do everything it can to ↑ retention of fluid (because it thinks blood volume is low). The way it accomplishes this is by ↑ reabsorption of urea and sodium, since water will follow. This is why blood *urea* nitrogen is high (i.e., BUN/Cr >20) and sodium in the urine is low (FENa <1%). - Notice above I stress *sub-acute to chronic time frame* (i.e., days-months), since one of the highest yield points you need to know for USMLE is that *acute* ↓ perfusion (i.e., seconds to minutes) to the kidney from blood loss, acute heart failure exacerbation, or arrhythmia (i.e., VFib for 30 seconds before resusc) **causes acute tubular necrosis, not pre-renal.** - This is all over the NBME exams. For example, if they say a guy loses lots of blood during surgery and receives 20 packs of RBCs, then two days later while recovering in hospital he gets oliguria + deterioration of renal function, the answer is acute tubular necrosis, not pre-renal. Student says, "Wait, but there was ↓ renal perfusion though, so how does that make any sense. Isn't pre-renal caused by ↓ perfusion?" → *The PCT of the kidney is the most susceptible to anoxic/hypoxic injury due to the ↑concentration of ATPase transporters.* So acute drop in blood flow → acute hypoxia → PCT sheds. - There is a 2CK NBME Q where they mention a guy post-op who had *not* lost any blood during surgery + had *not* experienced any episodes of low BP + was started on ketorolac (an

	NSAID) for pain post-op + now has oliguria and ↑ creatinine → answer = "hypoperfusion" as cause; this is example of where NSAIDs, not ATN, can cause post-op oliguria from pre-renal. - HY causes of pre-renal for USMLE are NSAID or diuretic use, chronic left heart failure, or dehydration from days of vomiting/diarrhea. - USMLE will give patient who's been on an NSAID (e.g., naproxen) for several weeks, or who was commenced on furosemide (loop diuretic) a few days ago. - Vignette can give patient who is on an NSAID + now has peripheral edema + they ask why there's edema → answer = "↓ renal excretion of sodium." This is because the PCT ↑ reabsorption of sodium in pre-renal (FENa <1%). They will also sometimes give NSAID + edema + ask what you do, and the answer is just "discontinuation of ibuprofen."
Intra-renal	- Essentially synonymous with acute tubular necrosis on USMLE. - The kidney can't reabsorb urea and sodium as easily, so BUN is lower (i.e., BUN/Cr is <20) and sodium is higher in the urine (i.e., FENa is >1%). - I discuss acute tubular necrosis in more detail in its own table below, but as I mentioned above in red font, the most important point for USMLE is that you remember *acute* drop in perfusion to the kidney causes acute tubular necrosis, *not* pre-renal. I need to be an asshole and inculcate that.
Post-renal	- Almost always due to BPH on USMLE, but can also be due to ovarian or cervical cancer impingement on the ureter(s). - What you need to know is: old dude + high creatinine = BPH till proven otherwise. - The answer on USMLE can often just be "increased Bowman capsule hydrostatic pressure." - Next best step is "insertion of catheter" to relieve the urinary retention, even if the patient has bacteriuria (i.e., choose catheter insertion over antibiotics). - Another 2CK Q has "check post-void volume" as answer in elderly male with high Cr. USMLE won't force you to choose between catheter insertion of checking post-void volume. - Normal post-void volume is <50ish mL. If the USMLE wants overflow incontinence, they'll give post-void volume ~300-400+ mL. - Diabetic neurogenic bladder (i.e., hypotonic bladder from neuropathy to the detrusor muscle) can also lead to overflow incontinence and post-renal azotemia, but as I said, most Qs on USMLE will focus on the old dude with BPH. - Hydronephrosis can be seen as large, dilated kidneys in patients with obstruction, but this is rare diagnosis.  - Step 1 wants you to know above gross specimen post-nephrectomy in patient with Hx of hydronephrosis. They can give, e.g., kid with chronic ureteral obstruction + show above pic.

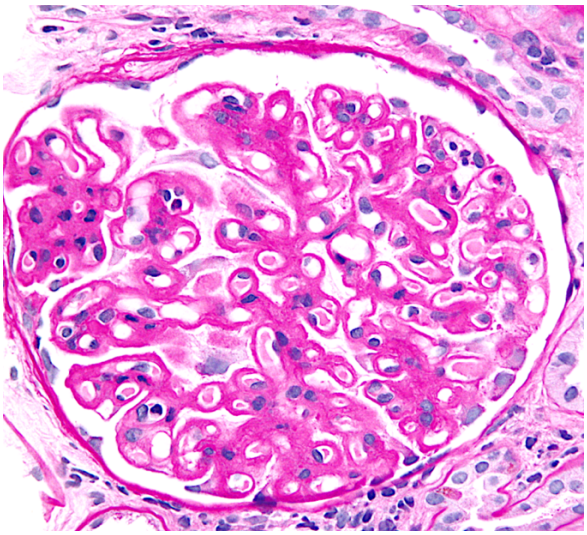
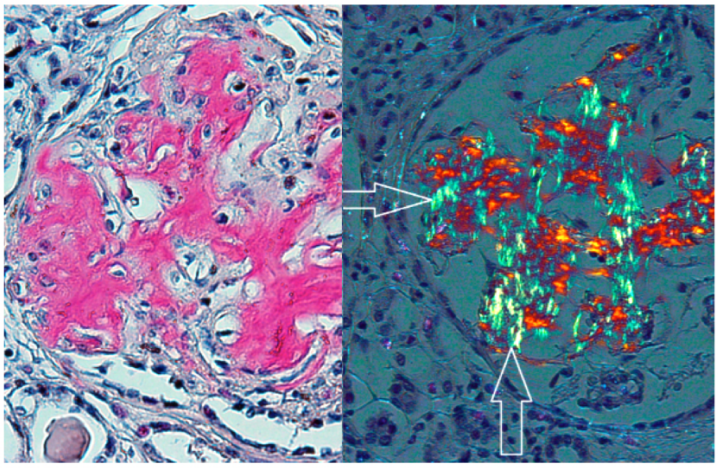
Nephritic vs Nephrotic syndromes
- Nephritic conditions have blood in the urine; nephrotic syndromes don't. - You need to memorize for USMLE which HY conditions are nephritic vs nephrotic. Don't worry, I keep things real clean and concise in the tables below without the bullshit nonsense of other resources. - There are four main points you should memorize as being part of each syndrome type. **Nephritic syndromes:** 1) **Hematuria** (blood in the urine). 2) **Oliguria** (↓ urinary output; the definition is <400 mL/day, but USMLE won't assess that number). 3) **Azotemia** (↑ blood urea nitrogen; this is because urinary output is ↓). 4) **Hypertension** (↑ RAAS due to inflammation of renal microvasculature). **Nephrotic syndromes:** 1) **Proteinuria** (nephrotic level by definition is >3.5g/day, but USMLE won't assess that number). 2) **Hypoalbuminemia** (due to the proteinuria; will be <3.5 g/dL, but USMLE doesn't care about the #). 3) **Peripheral edema** (due to the hypoalbuminemia → ↓ serum oncotic pressure → transudation of fluid into interstitial spaces; if severe can present with ascites; if the stem says "pre-sacral edema," this is nephrotic syndrome till proven otherwise on USMLE). 4) **Hyperlipidemia** (liver pumps out apolipoproteins in an attempt to preserve serum osmolality / oncotic pressure). - Should be noted that nephritic syndromes often have proteinuria, just usually not at nephrotic levels. Students sometimes erroneously think nephritic syndromes don't have proteinuria because it's not part of the 4-point categorization above. - You don't have to worry about the notion of which conditions are "both nephritic and nephrotic." What is most important is: simply know which conditions have hematuria and which ones don't (via tables below). - Nephrotic syndromes can ↑ risk of DVT, renal vein thrombosis, and varicocele due to loss of antithrombin III in the urine → hypercoagulable state. This is because nephrotic syndrome usually entails non-specific massive protein loss due to loss of size and charge barrier, and antithrombin is a protein. The NBME wants you to know left renal vein thrombus accretion from hypercoagulable state can cause varicocele.

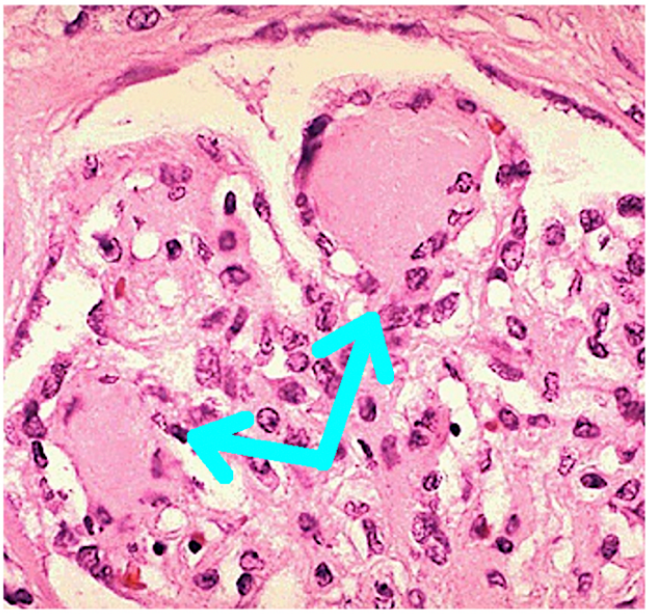
HY Nephritic syndromes	
PSGN	- Post-streptococcal glomerulonephritis; aka "proliferative glomerulonephritis"; the latter is asked on NBME, where they give vignette of PSGN, and answer is just "proliferative glomerulonephritis." Student is like what the hell? - Answer can sometimes be written by NBME as just "acute glomerulonephritis." - The answer on USMLE for red urine 1-3 **weeks** following a sore throat caused by Group A Strep (*Strep pyogenes*). This is in contrast to IgA nephropathy (which I discuss in detail below), which is red urine **1-3 days** following a sore throat (viral infection). - PSGN can be caused by skin infections. This can be impetigo (school sores), cellulitis, or erysipelas (I discuss these in detail in my HY Derm PDF). For example, the Q might say a 10-year-old has yellow crusties on his arm for the past 7 days (impetigo) + now has red urine → answer = "acute glomerulonephritis." - Type III hypersensitivity (antigen-antibody complexes that form in blood and deposit in the kidney). Don't confuse with rheumatic heart disease, which is a type II HS. - Serum complement protein C3 can be ↓ (can also be ↓ in SLE, but unrelated). - Streptolysin O or A titers will be ↑. - PANDAS is tested on Psych forms for 2CK → Pediatric Autoimmune Neuropsychiatric Disorder Associated with *Streptococci* → presents as new-onset tic/Tourette, OCD, or ADHD within weeks of Group A Strep infection. - Descriptors such as subepithelial deposits or "lumpy bumpy" appearance on electron microscopy (EM), etc., have been parroted across resources over the years but are essentially garbage for USMLE. - PSGN usually self-resolves in kids without sequela. But USMLE wants you to know that ↑ age means worse prognosis, where chance of renal failure is ↑ it if occurs in

	adults. This is probably related to the more robust immune response resulting in more advanced renal damage.
IgA nephropathy	- Aka Berger disease; IgA deposition in renal mesangium. - Red urine 1-3 **days** after a sore throat. This is in contrast to PSGN, which is red urine 1-3 **weeks** after a sore throat. I just mentioned it above obviously. But students fuck this up despite the inculcation so I'm reiterating it like an asshole. - Caused by **viral infection,** not Group A Strep. - Can sometimes be caused by GI infections, but USMLE usually avoids this etiology unless including it in the Henoch-Schönlein purpura constellation. - Henoch-Schönlein purpura is LY for Step 1 but HY for 2CK Peds. It's a tetrad: 1) Palpable purpura (usually on buttocks/thighs). 2) IgA nephropathy (red urine). 3) Arthralgias. 4) Abdominal pain. - All 4 need not be present for HSP, but the abdo pain component here is presumably viral gastroenteritis leading to IgA nephropathy.
Goodpasture syndrome	- The answer on USMLE for a male 20s-40s with hematuria and hemoptysis. Similar to Wegener and microscopic polyangiitis, can cause RPGN. - Caused by anti-glomerular basement membrane antibodies (anti-GBM), which are antibodies against collagen IV. - Mnemonic: "2,3,4. 2,3,4. The goodpasture is marching through the field. 2,3,4." → Type 2 hypersensitivity against the alpha-3 chains of type 4 collagen. - Renal biopsy will show linear immunofluorescence. USMLE likes showing the bright green image for this.  - Do not confuse this with Alport syndrome, which is an XR condition caused by mutations in collagen IV.
Alport syndrome	- X-linked disease; *mutation in* collagen IV gene. - Do not confuse with Goodpasture syndrome, which is antibodies against type IV collagen. - The answer on USMLE for a male who has red urine + an eye or ear problem (collagen IV is present in basement membranes in the kidney, ear, and eyes). - The "eye problem" can be blurry vision/cataracts; the "ear problem" will be neurosensory hearing loss (due to organ of Corti dysfunction). - Some students have asked if it's XR or XD. Literature appears to be varied, but USMLE doesn't care. On offline NBME 18, however, it's mentioned in a Q as XR. - You can memorize "splitting of the lamina densa" as associated with Alport.
MPGN	- Membranoproliferative glomerulonephritis. - The answer on USMLE for red urine in a patient with hepatitis C, heroin use, or malignancy.

	- Descriptors such as “dense deposits,” “C3 nephritic factor” and “duplication of basement membranes” are all basically nonsense for USMLE. - Answer is “renal biopsy” as the next best step. This guides our management. - A 2CK IM form has “heroin” as the answer (heroin-induced nephropathy) for patient with protein and blood in the urine. In contrast, if heroin causes a renal issue without blood, the answer they want is FSGS.
DPGN	- Diffuse proliferative glomerulonephritis. - The answer on USMLE for red urine in a patient with SLE. - Can sometimes be associated with “wire-looping of capillary walls.” - Same as with MPGN, USMLE wants “renal biopsy” as the next best step, since this guides our management.
RPGN	- Rapidly progressive glomerulonephritis. - Acute deterioration in renal function in the setting of a vasculitis (i.e., Wegener, microscopic polyangiitis) or Goodpasture syndrome. I discuss these conditions in detail later in this PDF, don’t worry. - Characterized by **fibrin crescents** on biopsy. 

HY Nephrotic syndromes	
Minimal change disease	- The answer on USMLE for otherwise unexplained edema in a kid who doesn’t have blood in the urine. - Presents as peripheral edema, periorbital edema, and/or ascites. - Almost always this is pediatric. Very rarely it can be due to Hodgkin in adult. - Classically post-viral (i.e., URTI), but ~50% of vignettes won’t mention that. - In other words, textbook vignette is an 8-year-old who has the sniffles for 4 days, followed by peripheral and periorbital edema a week later, without blood in the urine. Once again though, the stem need not mention the viral infection. - Called minimal change disease because light microscopy (LM) shows no abnormalities. EM, however, shows effacement of the podocytes. - MCD is aka lipoid nephrosis (lipid droplets can be seen in urine on LM). - Mechanism for nephrotic syndrome is “loss of size and charge barrier.” - Corticosteroids are the treatment and are highly effective. - A 2CK Peds Q gives minimal change disease as etiology for spontaneous bacterial peritonitis (i.e., any cause of ascites can cause SBP; I discuss this stuff in the GIT PDF in extensive detail).
Membranous glomerulonephritis / nephropathy	- The answer on USMLE for nephrotic syndrome in patient who has: - Exposure to drugs: dapsone, gold salts, sulfonamides. - Infection: hepatitis B (Hep C can cause it too, but rare). - Visceral cancers, e.g., breast, pancreatic.

	- Autoimmune (primary): antibodies against phospholipase A2 receptor. - The LM image is important and shows highly eosinophilic (pink) and inflamed / thickened capillary walls.  - Can cause "spike and dome" appearance of subepithelial deposits on EM.
Renal amyloidosis	- The answer on USMLE for nephrotic syndrome in multiple myeloma. - Amyloidosis means protein depositing where it shouldn't be depositing. - Multiple myeloma = cancer of plasma cells that produce too many immunoglobulins → high serum IgG kappa/lambda light chains. Immunoglobulins are proteins → these fly through the kidney and cause Bence-Jones proteinuria; they also deposit in the renal parenchyma, causing renal amyloidosis. - Biopsy shows classic apple-green birefringence with Congo red stain.  - For some magical reason, USMLE cares you know that the amyloid appears this way because it is β-pleated sheets (i.e., answer on NBME; α-helices is wrong); makes sense since the former are "flat" and can reflect light at different angles. - An NBME Q has "renal parenchymal disease" as answer for kidney issue due to multiple myeloma.
Diabetic glomerulosclerosis	- Diabetes is the most common cause of chronic renal failure. - Renal failure simply means ↓ glomerular filtration rate (GFR). - USMLE wants the first two changes that occur in the kidney due to diabetes as follows: 1) Hyperfiltration (↑ glucose crossing the glomerular basement pulls water with it → polyuria). Sometimes the Q can just simply give you diabetes +

	they ask what's most likely to be seen initially → answer = "↑ glomerular filtration." Over time, the kidney will fail and GFR will ↓. 2) Thickening of the glomerular basement membrane (non-enzymatic glycosylation of the membrane → loss of size and charge barrier). - Kimmelstiel-Wilson nodules are composed of hyaline and look like pink circles on light microscopy. This image is very HY.  - Hyaline arteriolosclerosis is deposition of hyaline (protein) in the walls of the renal vasculature. Although this can be due to many things, on USMLE, they want you to know this is caused by diabetes. - ACEi (e.g., lisinopril) or ARBs (e.g., valsartan) are the first drugs used in diabetes if there is HTN (>130/80 in diabetes, not 140/90), proteinuria, or ↑ creatinine or renin. - AT-II normally constricts efferent arterioles leaving the kidney, causing a backup of pressure at the glomerulus. This serves the purpose of maintaining GFR (i.e., filtration through Bowman capsule) when renal blood flow has ↓. - In other words, when renal perfusion ↓, RAAS goes ↑, AT-II goes ↑, and we get constriction of efferent flow leaving the kidney → ↑ filtration fraction (GFR/renal plasma flow), where we have GFR staying the same despite ↓ perfusion. - ACEi/ARBs ↓ constriction of efferent arterioles (they don't "dilate"; this refers to the action of prostaglandin on the afferent arterioles going to the kidney), thereby ↓ filtration fraction and reducing GFR. This might sound like a bad thing – i.e., "Why would we want to ↓ GFR in diabetes if it's the most common cause of chronic renal failure?" → The answer is because the slightly reduced GFR and filtration fraction early on ↓ the rate at which excess glucose is filtered across the glomerulus. Over time, it is inevitable that the glomerular basement membrane will non-enzymatically glycosylate and thicken, so ACEi/ARBs slow this process (i.e., slightly reduce GFR early in order to prevent a massive decrease later).
Focal segmental glomerulosclerosis (FSGS)	- The answer on USMLE for nephrotic syndrome in sickle cell, heroin, or HIV. - Not as responsive to steroids as MCD. - FSGS literally means "some parts of some nephrons are affected." - Focal = some glomeruli. - Diffuse = all glomeruli. - Segmental = part of nephron. - Global = all of nephron.

Acute tubular necrosis	
- Sloughing of PCT tubular epithelial cells due to chemical or ischemic insult. - Only ~50% of Qs will mention "muddy brown granular casts," or "dirty brown granular casts," or "brown, pigmented casts." - The other ~50% of Qs will just mention oliguria or acutely worsening renal function in patient with classic ATN etiologies. Next best step in diagnosis is "measurement of urine sodium and creatinine concentrations." We expect FENa >1% and BUN/Cr <20 almost always for ATN.	
Cause	**HY points**
Drugs	- Can be caused by drugs: **aminoglycosides** (gentamicin, tobramycin, amikacin); cisplatin (a chemo agent), or IV contrast. - Any of the above drugs + new-onset oliguria or ↑ creatinine (there need not be mention of casts) = ATN. - For instance, NBME simply tells you patient was given many drugs and now has creatinine of 2 (NR 0.1-1.2) + no mention of casts + they ask for what was given → answer = gentamicin. Not hard. - Patient who's recently been treated for endocarditis + now has new-onset oliguria = ATN due to gentamicin (empiric Tx for endocarditis = gentamicin + vancomycin [new literature suggests vancomycin is not reliably nephrotoxic]). - Contrast nephropathy is exceedingly HY on 2CK, where they want you to know that **adequate hydration** prior to IV contrast is the number-one way to prevent it. For instance, they'll say patient got a CT + now has ↑ creatinine + they'll ask how it could have been prevented → answer = "0.9% saline," or just "IV hydration therapy."
Rhabdomyolysis	- Myoglobin is nephrotoxic. Big risk factors are **alcoholism, falls,** intense training (e.g., marathon), McCardle syndrome, statins/fibrates. - **Rhabdo causes a false (+) blood on urine dipstick.** They will say urinalysis shows 2+ blood but only 1-2 RBCs/hpf. This is because the dipstick can't differentiate between free myoglobin and hemoglobin on RBCs. - One NBME Q gives 3-4 RBCs/hpf as negative. I've seen this fool some students who were already aware of the false (+) blood point about rhabdo. If the Q wants (+) RBCs, they'll say like 10-20+/hpf. - The other half of rhabdo vignettes won't mention the false (+) urinalysis and will instead just say oliguria. You need to know that refers to ATN. - For example, an NBME-favorite vignette is patient being found at bottom of stairs in his/her house + has false (+) blood on urine dipstick and/or oliguria → answer = rhabdomyolysis. - They also like ↑ K^+ as part of rhabdo vignettes (makes sense if we have renal failure from ATN) and ↑ serum creatine kinase (CK). - Q can say 44-year-old guy found on bench in park has ↑ K^+, ↑ CK, and oliguria → diagnosis = rhabdo. Or they'll say just ↑ CK + oliguria, and they ask what is most likely to be seen in this guy → answer = ↑ K^+.
Acute ischemia	- It's to my observation **acute ischemia** is the highest yield cause of ATN on NBME exams. - Even though the medulla of the kidney technically receives less blood flow compared to the cortex, as I mentioned earlier, the USMLE is obsessed with the PCT of the kidney as most susceptible to anoxic / hypoxic injury. This is due to the ↑ concentration of PCT ATPase transporters, so acute drop in blood flow → acute hypoxia → PCT sheds. - Step 1 Qs love *cellular swelling* as part of this mechanism – i.e., acute ischemia → ↓ activity of PCT ATPases → buildup of intracellular Na^+ → water stays with sodium → tubular cell swelling. Answer on NBME for why there's swelling simply = ↓ ATPase activity (this is similar mechanism for why we get swelling + hemolysis in pyruvate kinase deficiency). - Acute ischemia to the kidney can be due to loss of blood from trauma/surgery, acute arrhythmia (e.g., 30 seconds of Vfib), MI, or **acute** exacerbation of heart failure (on 2CK Free 120). **These conditions cause ATN, not pre-renal.** Once again, ↓ flow to

	the kidney causing pre-renal azotemia will be more subacute/chronic, such as due to NSAID use, recent initiation of diuretic, or dehydration. There are rare Qs that are exceptions, but that is the general principle. - As mentioned earlier, USMLE will give a one-liner where they say, "Dude had surgery where he lost a lot of blood + received many packs of RBCs. 2 days later, he's now recovering in hospital + gets oliguria" (they don't say anything about brown casts) → answer = ATN, not pre-renal. - Likewise, they'll say woman had surgery + had 30-second episode intra-operatively where BP fell to 80/40 → answer = ATN, not pre-renal. - Guy has an MI (cardiogenic shock) + new-onset oliguria → ATN, not pre-renal. - Guy has burns covering 50% of his body + develops oliguria → ATN due to excessive fluid loss → acute ↓ perfusion to kidney. - Acute exacerbation of heart failure causing ATN is a difficult one since heart failure is classically associated with pre-renal from chronic ↓ blood flow to the kidney. But in acute exacerbation of heart failure, we have acute ↓↓ blood flow, leading to ATN (as I said, it's on Free 120).

Other renal conditions associated with ischemia	
Diffuse cortical necrosis	- Diffuse cortical necrosis is a term thrown around sometimes and causes confusion for students. I don't think I've ever seen USMLE assess this. I've only seen it show up rarely as an incorrect answer choice. - DCN is simply considered an extension of ischemic ATN, where DCN can occur in severe cases of ischemia (e.g., obstetric hemorrhages). - You can be aware it exists, but don't worry about it for USMLE.
Renal papillary necrosis (RPN)	- Renal papillae = parts of kidney where the collecting ducts meet the ureters. - The answer on USMLE for red/dark urine in a patient with sickle cell. - In contrast, if a sickle cell patient has a renal condition + no blood in the urine, the answer = FSGS, as mentioned earlier. - RPN is loss of the renal medulla from ischemia, but unlike ATN, the etiology is usually more subacute or chronic. **NSAIDs** are a notable cause in this regard, where ↓ afferent blood flow due to ↓ vasodilating prostaglandins over weeks-months causes the medullar blood flow to be slowly choked off. - RPN can occur more acutely in sickle crises, where sickling within microvasculature feeding the medulla results in focal pockets of ischemia. - Pyelonephritis (discussed later) can also cause RPN due to inflammatory compression of the medullar microvasculature. There's one NBME Q I can recall where renal papillary necrosis is the answer due to infection. But I'd say, overall, the highest yield point is just remembering red/dark urine in sickle cell = RPN.

Interstitial nephritis
- Exceedingly HY renal condition often confused with ATN. - Aka interstitial nephropathy, or tubulointerstitial nephritis/-nephropathy. - Think of this as "an allergic reaction of the kidney." - 4/5 Qs will be an **NSAID, β-lactam, cephalosporin, or allopurinol** followed by getting a maculopapular rash and WBCs (eosinophils) in the urine. This presentation is textbook/pass-level. - Only ~50% of vignettes will mention the maculopapular rash. - The stem need not say "eosinophils" either. They can just say patient is on β-lactam + now has WBCs in the urine + no mention of rash → answer = interstitial nephritis. If they say the rash, it's even easier. - 1/5 Qs will just say a patient was on an NSAID, β-lactam, or cephalosporin and now has mild proteinuria and hematuria. They don't mention a rash or WBCs in the urine. - There is also a singular NBME Q where they just say a patient has simple peripheral edema due to an NSAID, with no mention of anything else, and the answer is interstitial nephropathy. This is more unusual,

since “NSAIDs + edema” classically = pre-renal, but I’ve seen the NBME assess interstitial nephropathy for this as well.
- NBME also can give you simple vignette of interstitial nephritis, and then ask for the location in the kidney that’s affected (e.g., efferent arterioles, etc.) → answer = “renal tubule.”
- For example, 40M + treated with nafcillin for 6 weeks for MSSA endocarditis + now has maculopapular rash and eosinophils in the urine; diagnosis? → interstitial nephritis.
- 60F + using naproxen (an NSAID) for 6 weeks for her osteoarthritis + has peripheral edema → answer = interstitial nephropathy.
- 35F + just recently finished 10-day course of cephalexin + has mild proteinuria and hematuria → answer = interstitial nephropathy.
- 35F + taking ibuprofen + has maculopapular rash and eosinophils in the urine; Q asks where in the kidney is fucked up → answer = “renal tubule.” Student says, “But don’t NSAIDs affect the afferent arteriole?” → Yes, but this particular presentation is clearly tubulointerstitial nephropathy. I haven’t seen them be ambiguous here with “NSAID + peripheral edema alone,” where you have to debate whether it’s interstitial nephropathy or pre-renal. They’ll either ask one or the other.

Renal tubular acidosis

Quick notes before discussing RTA:

MUDPILES = mnemonic for high anion-gap metabolic acidoses = **M**ethanol, **U**remia (renal failure), **D**KA, **P**henformin (a drug you don’t have to worry about), **I**ron/**I**soniazid, **L**actic acidosis, **E**thylene glycol, **S**alicylates (aspirin).

Anion-gap is calculated as Na^+ - (Cl^- + HCO_3^-). Normal range is 8-12. High anion-gap = 13 or greater.

- USMLE really doesn’t give a fuck that you know the specifics of types I, II, and IV.
- The way they assess this is by you knowing this is a type of **normal** anion-gap metabolic acidosis (i.e., it is not part of MUDPILES).
- For example, you’ll get a 15-line massive paragraph + tons of lab values + have no idea what’s going on, but then you calculate the anion gap as 12 (NR 8-12), so you can eliminate all of the MUDPILES answer choices, such as lactic acidosis, DKA, and ethylene glycol poisoning, and you’re left with, e.g., Crohn disease or renal tubular acidosis. Then you just say, “Well this clearly ain’t Crohn, so it must be RTA.” That is how I would say 4/5 RTA Qs show up.

- RTA type I = ↓ ability to secrete H^+ in the cortical collecting duct.
- RTA type II = ↓ ability to reabsorb HCO_3^- in the PCT.
- RTA type IV = renal resistance to aldosterone, or hyporeninemic hypoaldosteronism.

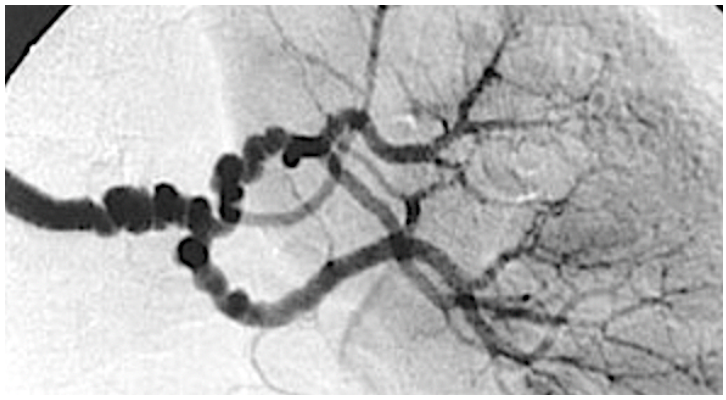
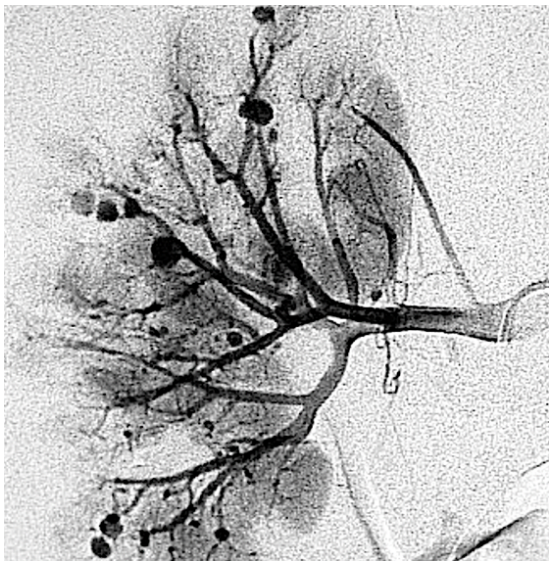
- Type IV presents with hyperkalemia. The others do not. Type III apparently is hyper-rare.
- Type IV will be a patient who has Addisonian-like picture (i.e., ↓ Na+, ↑ K+, ↓ HCO_3^-), but the vignette will present in a patient who has chronic renal failure.
- Type II can be caused by Fanconi syndrome (discussed more in table below).
- Type I can apparently have renal parenchymal stones (i.e., stones in the actual tissue of the kidney, rather than within the tubular lumina as with traditional nephrolithiasis).
- As I said, the key point is you just know RTA is normal anion-gap / not part of MUDPILES. That’ll cover you like 4/5 times.

Other renal conditions

Tumor lysis syndrome	- Lysis of tumor cells (especially leukemic) by chemoradiotherapy leads to release of uric acid and phosphate that can overwhelm the kidneys and cause crystals, namely from uric acid. - USMLE loves asking what can be given to prevent renal failure in patient undergoing chemo → answer = “allopurinol.” - Or they’ll ask for MOA of drug that can prevent renal failure → answer = “xanthine oxidase.”

	- The arrows USMLE wants for what will be seen in the blood in tumor lysis syndrome is same as renal failure except we also have ↑ uric acid.
Fanconi syndrome	- Inability of the PCT to reabsorb "lots of stuff." - Shows up on NBME as an arrow Q, where they ask about the reabsorption of amino acids, glucose, bicarbonate, and phosphate, and the answer is a ↓ for all of them. - Since bicarb reabsorption is impaired, this can lead to RTA type II, but USMLE doesn't really give a fuck. The key point is the ↓ for the reabsorption of the above substrates. - For whatever magical reason, an important cause of Fanconi syndrome is consumption of expired tetracyclines. - Don't confuse Fanconi syndrome with Fanconi anemia. The latter is an autosomal recessive aplastic anemia with hypoplastic thumbs/radii (I've made YouTube clips discussing this).
Hartnup disease	- Inability of the PCT to reabsorb tryptophan. - Can cause niacin (B3) deficiency in theory, since tryptophan is a precursor in niacin synthesis. - This is more just textbook masturbation to be honest that's been parroted over the years. Can't say I've seen this ever assessed once. Just mentioning it here otherwise I'll get some umbrageous student popping into my DMs about it.
Cystinuria	- Cystine stones (hexagonal) can occur in young adult due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystine is two cysteines put together. - Cystinuria can be diagnosed with cyanide-nitroprusside test.
Hepatorenal syndrome	- Nonsense diagnosis on USMLE that will sometimes show up as an incorrect / distractor answer choice. - Apparently liver failure can sometimes cause renal failure via obscure mechanisms. - Renal biopsy will be normal almost always.
Thin glomerular basement membrane disease	- Benign familial condition where patient can have mild hematuria. - Thin basement membranes allow for passage of small amounts of RBCs. - No treatment necessary.

Renovascular hypertension	
Renal artery stenosis (RAS)	- Narrowing of one or both renal arteries due to atherosclerosis that causes ↑ renin-angiotensin-aldosterone system (RAAS) and ↑ BP. - Q will be patient over the age of 50 with cardiovascular disease risk factors, such as diabetes, HTN, and/or smoking. - Patients who have pre-existing HTN causing atherosclerosis leading to RAS will often have 10-20 years of background HTN that then *becomes accelerated* over a few-month to 2-year period. What this means is: the slowly developing atherosclerosis in the renal arteries finally reaches a point at which the kidney is unable to maintain autoregulation, and the RAS is now clinical (i.e., accelerated HTN of ↑↑ BP within, e.g., 3 months). - Another way USMLE will give RAS is by giving ↑ BP in patient with significant evidence of atherosclerotic disease (i.e., Hx of coronary artery bypass grafting, intermittent claudication), and then ask for the most likely cause → RAS. You have to say, "Well he clearly has atherosclerosis in his coronaries and aortoiliac vessels, so that means he'll have it in the renal arteries too." - Q can say older patient with carotid bruit has recent ↑↑ in BP and then ask for diagnosis → answer = RAS. Similar to above, if the patient has atherosclerosis in one location (i.e., the carotids), then he/she will have it elsewhere too. - HY factoid about RAS is that **ACEi or ARBs will cause renin and/or creatinine to go up.** This is a HY point that is often overlooked and is asked on NBMEs. I have not seen NBME care whether it's uni- or bilateral in this case. → Kidney

	can autoregulate across flux in perfusion. Patients with already-compromised renal blood flow are more sensitive to the subtle ↓ in filtration fraction that occurs secondary to ACEi/ARB use, so renin/creatinine ↑. - If USMLE gives you unilateral RAS, renin is only ↑ from that kidney. The other kidney will not produce ↑ renin. - After renin and aldosterone levels are obtained, **MR angiography** of the renal vessels is what USMLE wants for the next best step in diagnosis.
Fibromuscular dysplasia (FMD)	- The answer on USMLE for narrowing of the renal arteries in a woman 20s-40s. - **Not** the same as renal artery stenosis, and **not** caused by atherosclerosis. - If you broadly say "renal artery stenosis," that specifically refers to atherosclerosis of the renal arteries in patient >50 with CVD. - FMD is tunica media hyperplasia (not dysplasia, despite the name) that results in a "string of beads" appearance on renal angiogram.  - MR angiography is answer on NBME for diagnostic modality. - Can affect the carotid vessels. A 2CK Surg Q gives FMD vignette and also says there is 25% occlusion of one of the carotids.
Polyarteritis nodosa (PAN)	- As discussed in the cardio PDF, this is a medium-vessel vasculitis that causes a "string of pearls" appearance of the renal vessels. This can be confused with FMD, but note the difference in the imaging. FMD shows more beading along the larger/proximal arterial sections, whereas PAN shows pearls more at the vascular termini, with the more proximal parts more likely to be spared.  Angiogram of PAN shows pearls that are more distal in the renal vasculature; FMD, in contrast, shows more proximal beading, with the termini not as conspicuously involved.

	- There is an NBME Q where they list both PAN and FMD as answer choices and it relies on you knowing the angiogram to get it right. They don't mention hepatitis B, but answer is PAN. That's why I'm harping on this here. - Causes fibrinoid necrosis. I've also seen this on NBME as "segmental, ischemic necrosis." - **Can be caused by hepatitis B.** - Spares the pulmonary vasculature.
Renal artery thrombosis	- The answer on USMLE for hypertension in a neonate following umbilical artery catheterization. USMLE simply wants "↑ renin, ↑ aldosterone" as the answer. - Sounds weird, but you need to know umbilical artery catheterization is a major risk factor for renal artery thrombosis in neonates. - 2CK NBME Q gives brief umbilical artery catheter insertion in kid born 26 weeks' gestation in order to monitor blood pressure for a pneumonia. 3 weeks later, he has ↑ BP (128/86) → answer = "↑ renin, ↑ aldosterone" as answer. - BP in term neonates should be ~60/40. In a preemie 29 weeks' gestation, it should be ~50/30 according to Google.

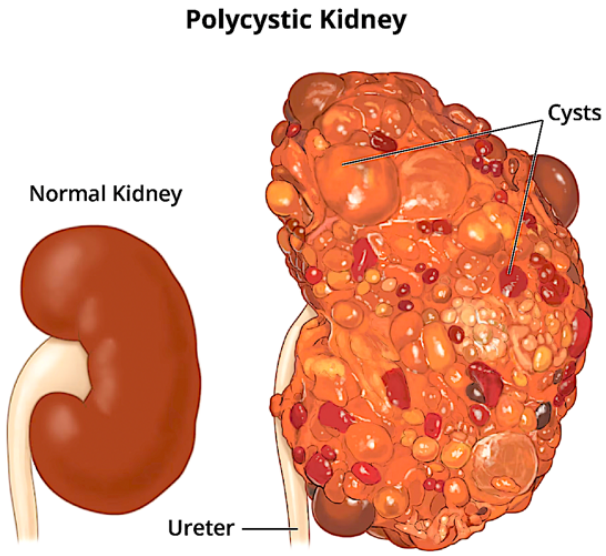
HY Vasculitides affecting the kidney	
- Vasculitides is a fancy word that is pleural for vasculitis.	
Granulomatosis with polyangiitis	- Formerly known as Wegener granulomatosis. - Answer on USMLE for adult with triad of 1) hematuria, 2) hemoptysis, and 3) "head-itis" – i.e., any problem with the head, such as nasal septal perforation, mastoiditis, sinusitis, otitis. - Associated with cANCA and anti-proteinase 3 (anti-PR3) antibodies. - Causes "necrotizing glomerulonephritis" that can lead to rapidly progressive glomerulonephritis (RPGN). - USMLE can give a vignette of Wegener and then the answer is just "vasculitis."
Microscopic polyangiitis	- Will just present as hematuria in a patient who is pANCA / anti-MPO (+). - Similar to Wegener, can cause RPGN.
- The above conditions can be associated with a weird neuropathy called mononeuritis multiplex, which means neuropathy of "one large nerve in many locations" – e.g., wrist drop + foot drop in same patient. - You don't need to worry about the fancy term "mononeuritis multiplex," but what I do want you to be aware of is that neuropathy will sometimes show up in these vasculitis vignettes, so don't be confused about it. - Eosinophilic granulomatosis with polyangiitis (formerly known as Churg-Strauss) less commonly has renal involvement compared to Wegener and MP. I discuss many vasculitides in more detail in the HY Cardio PDF.	

HY Uremia points	
- As mentioned earlier, "uremia" means "urea in the blood," but is used to refer to patients who have symptomatic renal failure with poor lab values. - On USMLE, uremic patients will have: ↑ K^+, ↓ HCO_3^-, ↓ Ca^{2+}, ↑ PO_4^{3-}. Na^+ is variable.	
Uremic pericarditis	- Friction rub in patient with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic encephalopathy	- Mental status change in patient with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic platelet dysfunction	- Nosebleeds and/or petechiae in patient with renal failure. - **Qualitative, not quantitative,** platelet problem. In other words, bleeding time is ↑, but platelet count is normal. - Mechanism is ↑ BUN causing impairment of platelet function. - Answer on USMLE can be written as "acquired platelet dysfunction." - Treatment = hemodialysis.

	- On 2CK NBME 10, Surg Q wants "initiation of dialysis" as answer in uremic patient pre-op. Improving RFTs will ↓ peri-op morbidity/mortality + ↓ risk of intra-op bleeding due to uremic platelet dysfunction.
Uremic frost	- Yellow skin sometimes seen in severe renal failure due to ↓ ability to excrete urea. Excess is excreted in sweat, precipitating out as white crystalline deposits (frost). - Can be associated with itchy skin (uremic pruritis).

Urolithiasis	
- Urolithiasis is broad, umbrella term that refers to both nephrolithiasis and ureterolithiasis. - Diagnosed with non-contrast CT of abdomen and pelvis. - KUB (X-ray of kidney, ureters, bladder) can be done but isn't as sensitive (although below I show an X-ray).	
Stone type	**HY points**
Calcium	- Most common type of stone. Can be calcium oxalate or calcium phosphate, although I've never seen USMLE once assess or give a fuck about phosphate stones. - Most young adults with idiopathic kidney stones will have "normocalcemia and hypercalciuria" – i.e., normal serum calcium but elevated urinary calcium. - USMLE will give you healthy male in his 20s with sharp pain in the flank or groin + RBCs in the urine → answer = urolithiasis. - **Crohn disease** and disorders causing fat malabsorption increase the risk for calcium oxalate stones (↑ fat retained in GI tract → ↑ chelation with calcium in GI tract → ↓ calcium available to bind oxalate → ↑ oxalate absorption by GI tract → ↑ urinary oxalate). - Primary hyperparathyroidism and malignancy causing hypercalcemia are HY etiologies. - Milk-alkali syndrome: ↑ calcium + ↑ bicarb + calcium stones. Textbook scenario is a patient taking too many antacids, but vignette will usually not mention this and it will simply be a diagnosis of exclusion (i.e., you eliminate to get there). - Ethylene glycol (Anti-Freeze) and hypervitaminosis-C can cause oxalate stones. - First step in prevention and treatment of stones is "adequate hydration." - **Thiazides** can be used to prevent recurrent stones by ↓ urinary calcium. NBME has a Q that asks why → answer = ↑ reabsorption of calcium. - In theory, alkalinization of the urine can help treat oxalate stones, and acidification of the urine can help treat phosphate stones.
Struvite	- Composed of ammonium magnesium phosphate. - Formed in ↑ pH in the presence of urease (+) bacteria. - *Klebsiella, Serratia,* and *Proteus* are HY causes. - These stones are large, ram-horn like. The image is HY.  - Apart from adequate hydration, Tx is acidification of urine with NH_4Cl + surgery.

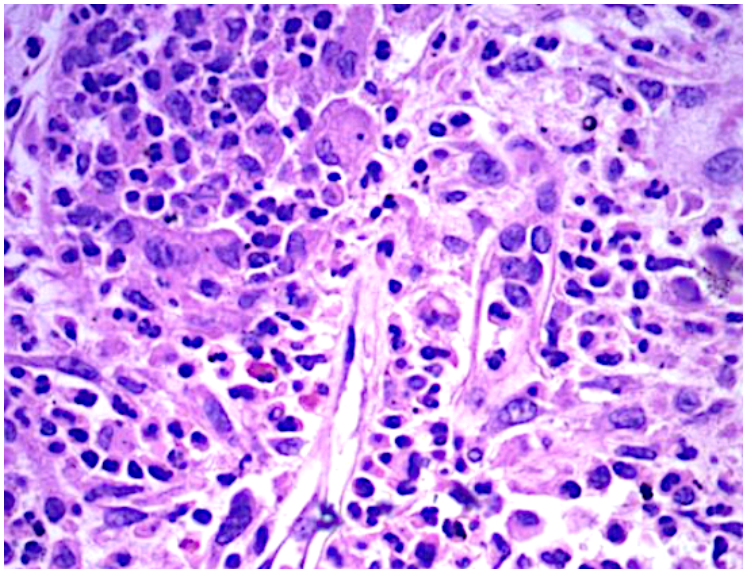
Uric acid	- Seen in patients who have gout. Not complicated. - First-line Tx for *chronic* gout is xanthine oxidase inhibitors (i.e., allopurinol or febuxostat). They are preferred over uricosurics (meaning ↑ urinary excretion of uric acid) like probenecid and sulfinpyrazone because the latter agents ↑ risk of uric acid stones. - Probenecid and sulfinpyrazone inhibit organic anion transporter (OAT), which normally functions to ↑ absorption of uric acid. - Weird factoid about probenecid is that it can help maintain serum β-lactam levels, since the latter are excreted by OAT. There's an NBME Q around somewhere that asks about this. Sometimes old-school docs will put a patient on a β-lactam and probenecid. Don't be confused by this. The patient doesn't have gout. It's just a way of ↓ renal excretion of the β-lactam.
Cystine	- As mentioned earlier, cystine is two cysteines put together. - Cystine stones (hexagonal) can occur in young adult due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystinuria can be diagnosed with cyanide-nitroprusside test.
Miscellaneous	- Acyclovir (for HSV and shingles) and indinavir (for HIV) can cause crystal nephropathy. Not particularly HY, but worthy of 1-2 lines here.

Cystic kidneys	
ADPKD	- Autosomal dominant polycystic kidney disease; chromosome 16. - The answer on USMLE if disease starts as an adult (i.e., 30s-40s). - Cysts are technically present early in life, but only become clinical as adult (i.e., ↑ BP and ↑ RFTs).  - These patients have ↑ BP due to ↑ RAAS (compression of microvasculature of kidney due to enlarging cysts). - Can cause saccular (berry) aneurysms of the circle of Willis → risk for subarachnoid hemorrhage. - Highest yield point is that **serial blood pressure checks** are correct over circle of Willis MR angiogram screening. Latter is wrong answer on USMLE. MR angiogram screening of circle of Willis is only done when there is (+) family Hx of SAH or saccular aneurysms. - Step 1 NBME gives easy vignette of ADPKD, and then the answer is just "polycystin" as the protein that's fucked up. Sounds obvious, but it's asked so I'm mentioning it. - Cystic kidneys are part of "ciliopathies," which is obscure term that refers to conditions where cilia are abnormal. Polycystin is a protein required for cilia function on renal epithelium. - Most common extra-renal location for cysts is the liver (85% by age 30).
ARPKD	- Autosomal recessive polycystic kidney disease; chromosome 6.

	- The answer on USMLE for cystic kidneys in pediatrics. - Can be associated with **hepatic fibrosis.**
Miscellaneous	- Just be aware renal cysts can occasionally develop in other conditions, such as chronic dialysis patients. - Conditions such as medullary sponge kidney are garbage for USMLE. Waste of time.

HY anatomic abnormalities	
Posterior urethral valves (PUV)	- Exceedingly HY for Peds for 2CK; also shows up on Step 1 NBME. - Most common genitourinary abnormality in neonatal males. - Urethra has abnormal presence of valves at the posterior (prostatic) urethra, preventing the outflow of urine. - USMLE will give Qs of varying severity. - Most severe is *in utero* oligohydramnios. - Can present as 12-hour-old neonate who hasn't yet urinated + has suprapubic mass (i.e., full bladder). - Can also present as 6-week-old boy who has full bladder (i.e., possibly slower accumulation due to only partially obstructed outflow). - Obstructed outflow ↑ risk for UTIs, cystitis, and pyelonephritis. - Apart from knowing this diagnosis, highest yield point is that we do **ultrasound followed by voiding cystourethrogram** to diagnose. - NBME Qs might already mention negative ultrasound in the vignette, or might omit it altogether, where you just select "voiding cystourethrogram."
Vesicoureteral reflux	- Congenital abnormality in which urine from the bladder can go back up into the ureter toward the kidney. - This ↑↑ risk for recurrent acute pyelonephritis → chronic pyelonephritis (as I discuss below). - Can also occur in pregnancy due to 2 reasons: 1) larger uterus in 3rd trimester can compress the ureters; 2) progesterone ↓ ureteral peristalsis. - This is why pregnant women can get pyelonephritis, and also why we always treat asymptomatic bacteriuria in pregnancy, whereas we don't treat it if woman is not pregnant.
Ureteral atresia	- Bit of a weird one, but not me being fancy. It's on NBME. - "Failure of canalization of proximal ureter" will be the answer if they tell you the renal collecting duct system is dilated, but the ureters are not dilated. - Sounds obvious, but I see students get this wrong a lot. - You need to simply know: kidneys → ureters → bladder → urethra, and if we have a congenital obstruction at any point, that could be referred to as "failure of canalization." - "Congenital ureteral obstruction" is an answer on one of the NBMEs for chronic pyelonephritis causing tubular atrophy (once again, discussed below).

Genitourinary infection-related stuff	
- For all of the STD/STI-related stuff, I cover that in the Repro/Obyn PDF. I'm keeping this small table more limited to kidney/bladder stuff.	
Acute pyelonephritis	- Infection of the kidney. "Pyelo" means kidney. - 9/10 questions will mention fever + **costovertebral angle (CVA) tenderness,** which is pain with percussion of the flank. - "Granular casts" can be seen. Do not confuse this with acute tubular necrosis. "General granular casts" can sometimes be seen in pyelonephritis and dehydration. It's on one of the 2CK IM forms, where students say, "Wait, but they say granular casts; how is this not ATN?" - Major risk factors are vesicoureteral reflux (especially in pregnancy) and posterior urethral valves.

	- Most common organism is *E. coli*. Other organisms, i.e., *Klebsiella, Serratia,* and *Proteus* cause struvite stones. - Neutrophilic infiltration will show up as lots of blue/purple (basophilic) cells on histo. USMLE likes this image.  - For above image, you say, "Mike I feel weird. Idk what I'm looking at." The blue cells are neutrophils infiltrating the kidney in acute pyelo. The USMLE will show images basically identical to this for a variety of infections, e.g., prostatitis, where the bigger picture concept is, "Oh that's acute inflammation. Those purple cells are neutrophils." *That's* what USMLE wants you to know. For instance, a nearly identical image of prostate histo in old dude with prostate pain and fever → you'd know immediately it's prostatitis. - It's to my observation **bacteria can be few in the urine** in acute pyelo. This confuses some students. But it's typically what I see on NBME forms. If the infection is further down, i.e., UTI in the urethra, then bacteria are more copious. - Treatment for pyelo is **ciprofloxacin** or **ceftriaxone.** USMLE is known to ask these. - Persistent fever despite Abx Tx → do CT to look for **perinephric abscess.** - For example, old dude + high Cr (caused by post-renal from BPH) + treated for pyelo → now gets sore ankle → was treated with cipro (causes tendonopathy). - 2CK form has ceftriaxone as an answer in pyelo Q where cipro isn't listed. But cipro is classic Tx. It's to my observation that ceftriaxone is HY drug on 2CK for community-acquired "general sepsis" or "general complicated/severe infections" – i.e., it is hard-hitting and covers wide array of community organisms. - There is one 2CK Q where they mention pyelo is treated with amp + gent in the stem, but I've never seen this assessed as an answer you need to choose. I've only ever seen ciprofloxacin or ceftriaxone as actual NBME answers for pyelo.
Chronic pyelonephritis	- Due to recurrent acute pyelonephritis. - Almost always pediatric, where they give a 4-year-old who has a small, shrunken kidney with **tubular atrophy** and **blunting and scarring of the renal calyces.**

	 Kidney appears to show loss of architecture + scarring. - USMLE will show you kidney that looks similar to the above in a child with recurrent UTIs, and then the answer is just "tubular atrophy" for what is most likely to be seen on microscopic examination. - Another Q shows the same image and the answer is "congenital ureteral obstruction" as the cause. - "Thyroidization of the kidney" is a buzzy phrase that has been applied to chronic pyelo over the years, but USMLE doesn't so much care about this. What they really like is the tubular atrophy and scarring / loss of architecture of the calyces. - Chronic pyelo doesn't present with cellular infiltration the way acute pyelo does. It refers to the kidney being destroyed/shrunken from repeated prior infections.
Cystitis	- Suprapubic tenderness in female. Patient need not have fever. - *E. coli* most common cause. HY for USMLE you know **fimbriae** and **pilus proteins** facilitate *E. coli*'s attachment to urothelium. - Will have bacteria and WBCs in the urine (i.e., bacteriuria + pyuria). - Urinary nitrites and leukocyte esterase can be positive. These just reflect bacterial infections. Some students get pedantic about which organisms result in which combo of (+) or (-) findings here, but that's low-yield for USMLE. - Can be caused by **suprapubic catheters** (on 2CK IM and Surg). - **Nitrofurantoin** is HY drug to treat cystitis on USMLE.
Chronic interstitial cystitis	- Not an infection. - This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care," and "physiotherapy."
General UTI	- Classically *E. coli* infection of the urethra. - Common in women due to shorter urethras. - As mentioned above for cystitis, **fimbriae** and **pilus proteins** facilitate *E. coli*'s attachment to urothelium. - Inoculation of the urethra following sexual activity is most common mechanism. - Can be caused by catheters. You need to know **dysuria + Hx of catheter = UTI.** - Tangentially, this is also HY for general sepsis, where if they mention Hx of IV line/catheter, USMLE wants you to be able to say, "Got it. That's the cause." - Trimethoprim, trimethoprim/sulfamethoxazole (TMP/SMX), or nitrofurantoin are standard treatments for UTI. - For prevention, post-coital voiding confers ↓ risk of recurrence.

	- If post-coital voiding doesn't work, "post-coital nitrofurantoin therapy" is answer on 2CK NBME form. - If post-coital nitrofurantoin therapy doesn't work, "daily TMP/SMX prophylaxis" is the answer. Sounds absurdly wrong / like a bad idea, but it's an answer a couple times on Obgyn assessments. - There's also a Q where they say a girl was treated successfully in the past with TMP/SMX for a UTI + ask how to prevent recurrent UTIs now, and they just jump straight to "daily TMP/SMX prophylaxis." - If patient is treated for a UTI and dysuria persists, the next best step is testing for *Chlamydia* and *Gonorrhea*.

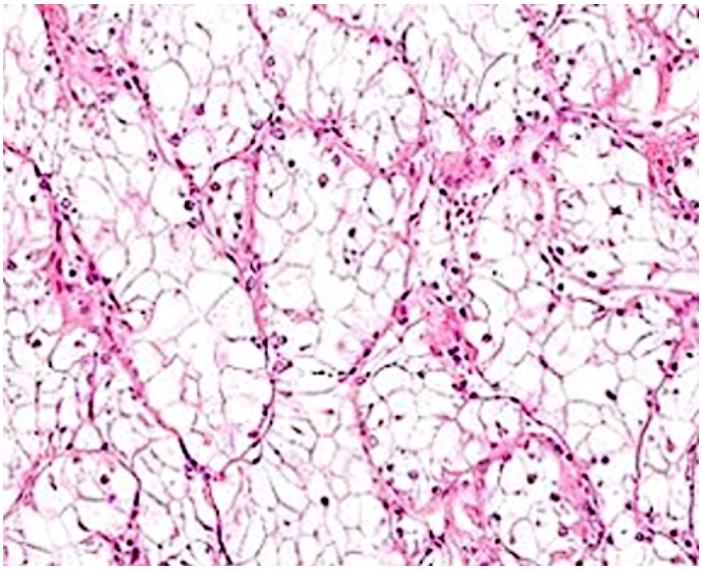
HY Bladder incontinences	
Stress	- The answer for loss of urine with ↑ intra-abdominal pressure from laughing, sneezing, coughing. - Stereotypical risk factor is grand multiparity (i.e., Hx of many childbirths) leading to weakened pelvic floor muscles. - I'd say only ~50% of Qs will mention Hx of pregnancy. The other ~50% are idiopathic. - Buzzy vignette descriptors and answer choices are: "downward mobility of the vesicourethral junction," "urethral hypermobility," and "urethral atrophy with loss of urethrovesical angle." - Treatment is pelvic floor (Kegel) exercises. These notably strengthen **levator ani, pubococcygeus,** and the **external urethral sphincter.** - USMLE is known to ask which muscle is *not* strengthened by Kegel exercises, which sounds obscure, since any muscle could theoretically be the answer ("Well the deltoid isn't strengthened."). But a favorite answer here is **internal urethral sphincter.** The way you know this is the answer is because internal sphincters are under sympathetic (i.e., involuntary; autonomic) control, which means it's impossible to strengthen it via a voluntary (i.e., somatic) exercise. USMLE doesn't expect you to be an obstetrician. The bigger picture concept is simply knowing internal sphincter control is involuntary. It is external sphincter control that is voluntary (somatic). - Do **not** give medications for stress incontinence on USMLE. - If Kegel exercises fail, patients can get a mid-urethral sling (LY; asked once).
Urge	- The answer on USMLE for patient who has an "urge" (NBME will literally say that word and I've seen students get the Q wrong) to void 6-12+ times daily **unrelated** to sneezing, coughing, laughing, etc. (otherwise stress incontinence). - Ultra-HY for **multiple sclerosis.** I've had students ask whether MS is urge or overflow. It shows up repeatedly on the NBMEs as urge; I've never seen it associated with overflow. I'd say ~1/3 of urge incontinence vignettes on NBME forms are MS. - Other vignettes will be peri-menopausal women, or idiopathic in old women. - Mechanism is "detrusor hyperactivity," or "detrusor instability." - Vignette can mention woman has urge to void when stepping out of her car, or when sticking her key in the car/front door of her house. Sounds weird, but these are important Qs to ask when attempting to diagnose urge incontinence. - UTIs can present similarly to urge incontinence. Some students have asked, "Well isn't that because UTIs are a cause of urge incontinence?" Not really. It just happens to be that UTIs can sometimes cause transient urinary urgency. For example, if they give a Q where they say patient had Hx of urinary catheter + now has dysuria and urinary urgency, answer = "urinary tract infection" on NBME; "detrusor hyperactivity" is wrong answer. - Treatment is **oxybutynin** (muscarinic receptor antagonist); this ↓ activity of the detrusor muscle of the bladder. - Some students get hysterical about mirabegron (β3-agonist), but I've never seen NBME forms assess this.
Overflow	- Will be due to either BPH or diabetes on USMLE.

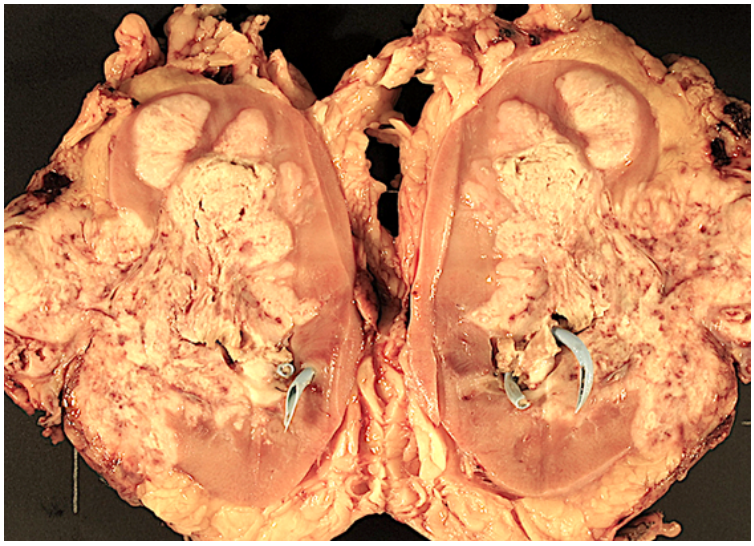
	- Will have ↑ **post-void volume.** Normal is < ~50 mL. On USMLE for overflow, they'll give you 300-400 mL as post-void volume. - As I talked about earlier for BPH, they will give old dude + high creatinine (post-renal azotemia). Next best step is "insertion of catheter" to relieve the obstruction. If they don't have this listed, "measurement of post-void volume" can be an answer. We then treat the BPH with finasteride (5α-reductase inhibitor) or an α1-blocker (tamsulosin, terazosin). - For diabetes, the mechanism is neuropathy to the bladder causing "neurogenic bladder," or "hypotonic bladder," or "hypocontractile bladder/detrusor muscle." - For neurogenic bladder causing overflow incontinence +↑ post-void volume, remember that USMLE is first obsessed with "measure post-void volume" and "insertion of catheter" if they are listed. They will not force you to choose between the two. But they like these answers prior to giving medications. - Give **bethanechol** (muscarinic receptor agonist); this stimulates the detrusor muscle. - Making sure you don't confuse oxybutynin and bethanechol is pass-level for USMLE.
Normal pressure hydrocephalus (NPH)	- Patient who has "wet, wobbly, wacky" presentation (i.e., urinary incontinence, ataxia, cognitive changes) +/- Parkinsonism (e.g., short-steppage gait). - Caused by failure of reabsorption of CSF by the arachnoid granulations, resulting in impingement on the zona radiata and **"failure to inhibit the voiding reflex"** (answer on NBME). - Enlargement of the lateral ventricles on head CT.
AIDS complex dementia	- Presents similar to NPH – i.e., "wet, wobbly, wacky" in AIDS patient.

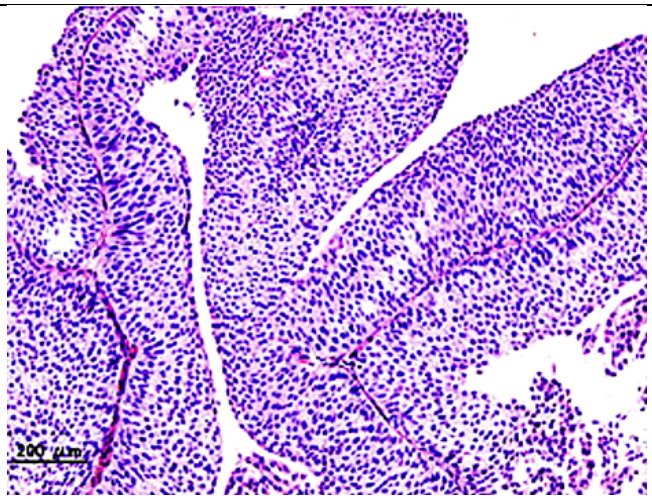
Urinary retention diagnoses	
Bladder neck dyssynergia	- Lack of coordination between bladder sphincters and detrusor muscle. - HY cause is spinal cord injury, which can cause either inability to void or urinary incontinence. - MS is another theoretical cause, which is where "overflow incontinence" as a result of MS has been asked by students, but once again, I've only ever seen urge incontinence for MS on NBME exams. Qbank ≠ NBMEs.
Bladder outflow obstruction	- Acute urethral obstruction due to **catheter removal** is HY on Surg forms. - They'll say patient has *anuria* (i.e., no urinary output at all) following removal of a catheter. - "Acute urethral obstruction" or "bladder outflow obstruction" is the answer. - Treatment on NBME is "re-insertion of urinary catheter."
Cauda equina syndrome	- Will present as leg pain + urinary retention on USMLE. - Can be idiopathic, from disc herniation, or other trauma. - Metastases is HY cause (i.e., random guy over 50; or female with Hx of mastectomy 5 years ago). - "Saddle anesthesia" is an overrated detail and doesn't show up in the vast majority of NBME Qs, as per my observation. - Cauda equina is the collection of nerves at the end of the spinal cord. Impingement on these nerves is what causes the syndrome. I cover this stuff in more detail in my HY Neuroanatomy PDF.

HY Genitourinary injuries	
Urethral injury	- Always presents with blood at the urethral meatus. - 3 ways this presents on USMLE: 1) Blood at urethral meatus + saddle injury (dude falls on balance beam). 2) Blood at urethral meatus + pelvic fracture/instability. 3) Blood at urethral meatus + "prostate not palpable."

	- Next best step is **retrograde urethrogram.**
Bladder injury	- Answer on USMLE for blunt force trauma to abdomen + exquisite suprapubic tenderness + catheter insertion yields 10-30 mL of dark bloody fluid. - Next best step is **retrograde cystourethrogram.** Do not confuse with voiding cystourethrogram, which is for posterior urethral valves.
Ureteral injury	- Just be aware this is USMLE-favorite injury in Obgyn for any surgery done in the gynecologic area. - Diagnosis is with retrograde pyelogram or CT urogram, although I haven't seen the USMLE give a fuck. Any ureter injury Q I've seen just has "ureter" as answer in the setting of a random gyn surgery.
Renal injury	- Exceedingly HY on both IM and Surg forms on 2CK. - Will present as bruising or pain over a flank following accident. - Blood in the urine is highly sensitive, meaning if it's not there, we can rule-out. - First step in diagnosis: "Is there gross blood in the urine?" If yes → do **CT of the abdomen with contrast** to diagnose (sounds wrong, since we're worried about kidney injury, so student says, "Why would we use contrast?" But it's what USMLE wants. **Ultrasound is wrong answer**). - If patient does not have gross blood in the urine, next best step is **urinalysis looking for microscopic blood.** If present → answer = CT of the abdomen with contrast. If negative → "no further diagnostic studies indicated."

Urothelial malignancies	
Renal cell carcinoma (RCC)	- Aka "clear cell carcinoma"; this is most common variant of RCC. - Classic Q is a smoker over 50 with red urine and a painful flank mass. - Q need not mention smoking, but it is biggest risk factor. - USMLE likes the histo for this, which will show you large clear cells.  - Can cause **polycythemia** due to ↑ EPO and **hypercalcemia** due to PTHrp secretion (correct, same as squamous cell carcinoma of the lung). - For example, they say 55-year-old male + red urine + polycythemia + hypercalcemia + show you above image; they ask for the source of the malignancy → answer = "kidney," not lung. Squamous cell carcinoma of the lung won't cause hematuria, nor does it metastasize to the kidney.
Wilms tumor (Nephroblastoma)	- The answer for painless flank mass in a kid age 2-4 years. - Does **not** present in newborns. If they give you a 12-hour-old male, for instance, with a midline mass, that is posterior urethral valves causing a full

	bladder (let alone the fact they say "midline," not flank). I've seen tons of students select Wilms for this, and I'm like, is the kidney in your midline? - If they give you a kid 2-4 years with a midline mass, that is neuroblastoma. - For whatever magical reason, you need to know Wilms tumor is sometimes caused by mutations on **chromosome 11.** They ask this factoid on 2CK as well. There are various Wilms tumor syndromes you need to be aware of: - **Beckwith-Wiedemann syndrome** - Wilms tumor + macrosomia (big baby >4,000g) + hemi-hypertrophy (half of the body is bigger than the other) + macroglossia + omphalocele + hypoglycemia. - Hard 2CK Peds Q gives newborn with macrosomia + hemi-hypertrophy + macroglossia + omphalocele; they don't mention a Wilms tumor (makes sense, since we said we won't see it in newborns); they ask for what else could be seen in this patient → answer = "hypoglycemia." I say hard Q because Wilms plays no role. **- WAGR syndrome** - **W**ilms tumor, **A**niridia (iris abnormalities), **G**enitourinary anomalies, **R**etardation. - Usually an easy Q, where they mention "aniridia" straight up in the vignette and then the answer is just "Wilms tumor" for what the kid can go on to develop. **- Denys-Drash syndrome** - Ambiguous genitalia + Wilms tumor.
Collecting duct carcinoma	- Rare as fuck. 0-3% of renal malignancies. But count on NBME asking it anyway.  - Offline NBME Q shows some pic of a kidney split open like this and then the answer is just "naphthylamine" (moth balls) as the causative agent.
Transitional cell carcinoma of the bladder	- Most common bladder cancer. - Classic vignette is hematuria in smoker *without* a painful flank mass or polycythemia/hypercalcemia (otherwise RCC). - USMLE wants you to know smoking is most common risk factor, but aniline dyes (industrial clothing dyes) and naphthylamine are important causes. - 2CK Surg Q gives Stage 0 bladder cancer vignette; answer is just "endoscopic resection." - I've never seen BCG vaccine as correct answer on NBME. I've only seen it as a distractor. For whatever reason, some studies in the past have suggested intra-cystic instillation of the TB vaccine can help with bladder cancer. But this is wrong on USMLE.

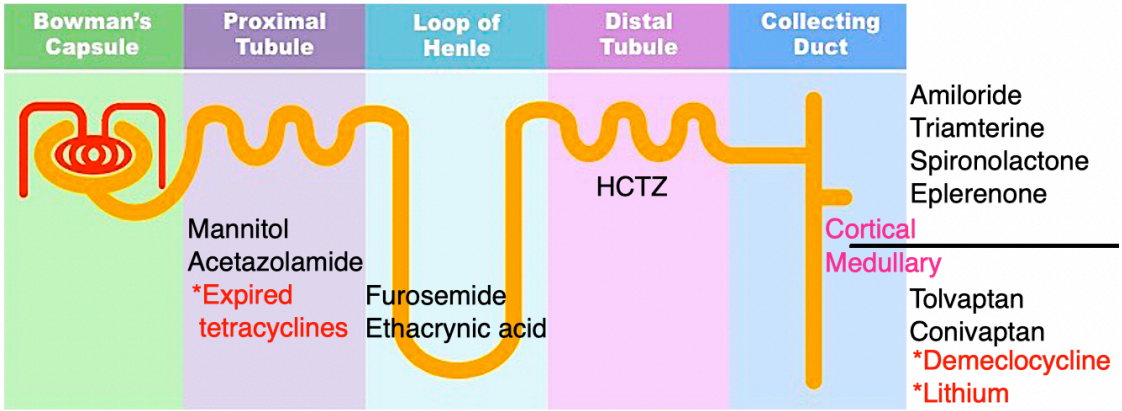
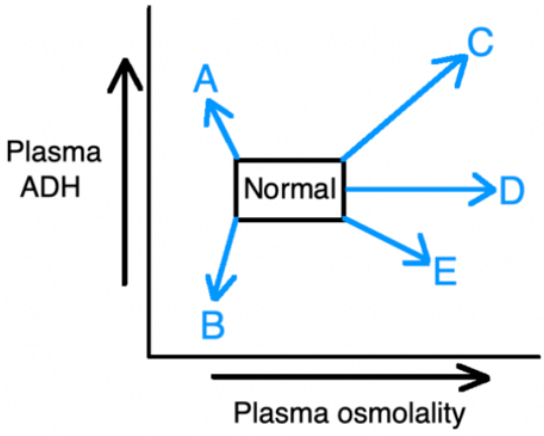
	 Can have papillary structure on histo.
Squamous cell carcinoma of the bladder	- Can be caused by *Schistosoma hematobium* (a trematode or fluke, which is a type of helminth). - *S. hematobium* lays its eggs in the bladder wall and cystic veins draining the bladder. - Vignette will be young guy who has red urine months after returning from Africa where he went swimming in a lake → answer is just *Schistosomiasis*. Or they ask what he's at ↑ risk of → answer = squamous cell carcinoma.

SIADH vs DI vs PP	
SIADH	- Syndrome of Inappropriate Anti-Diuretic Hormone secretion → means too much ADH (vasopressin secretion). - ADH is produced by supraoptic nucleus of hypothalamus → stored in posterior pituitary. - ADH ↑ free water reabsorption by the medullary collecting duct (MCD) of the kidney by causing aquaporin insertion. - Central SIADH → follows head trauma, meningitis, brain cancer, and pain (latter on 2CK Surg). - Ectopic SIADH → small cell lung cancer secreting ADH. - Drug-induced ADH → ultra-rare on USMLE, but carbamazepine can do it. - Patient will have dilute serum and concentrated urine: - ↓ serum sodium, ↓ serum osmolality, ↓ serum specific gravity. - ↑ urinary osmolality, ↑ urinary specific gravity. - You must know serum sodium is normally 135-145 mEq/L. So in SIADH, it's <135. - Serum vs urinary osmolalities will be all over the place and you do *not* need to memorize values. They might say serum osmolality is 250 and urinary is 750, and then you say, "Well I can tell serum is dilute compared to urine, which sounds like SIADH." - Specific gravity will be 1.000-1.030 on USMLE. Sounds obscure, but it shows up quite a bit, particularly on 2CK Qs. It's to my observation that values 1.000-1.006ish are "dilute"; 1.024-1.030 are "concentrated." For 9/10 Qs, values will be what you expect. - As I talk about in detail in my HY Arrows PDF, only the medullary collecting duct osmolality will change in response to ADH. The USMLE will ask you for the osmolality of the urine at different nephron locations in comparison to serum, and the combo is: **PCT isotonic; juxtaglomerular apparatus (JGA) hypotonic; MCD hypertonic.** → The PCT is always isotonic no matter what; the JGA (at top of thick ascending limb of loop of Henle) is always hypotonic no matter what; the MCD is clearly hypertonic in SIADH since we're pulling free water out of the urine. - "Fluid restriction" is first answer in diagnosis on 2CK. We want to see how serum/urinary values change in response.

	- Demeclocycline is answer on 2CK offline NBME 8 for treatment of SIADH. Demeclocycine is technically a tetracycline antibiotic but isn't used because it can cause insensitivity to ADH at the kidney (i.e., nephrogenic diabetes insipidus). So we essentially cause a 2nd problem that cancels out the 1st problem. - Conivaptan and tolvaptan are ADH receptor antagonists that can be used for SIADH.
DI	- Central diabetes insipidus → means not enough ADH secretion by hypothalamus, or the posterior pituitary is unable to release it properly. - Nephrogenic DI → insensitivity to ADH at the kidney (serum ADH is ↑). - Similar to central SIADH, central DI can be caused by head trauma, meningitis, and cancer. - Nephrogenic DI is caused by **lithium, demeclocycline, hypercalcemia,** and **NSAIDs.** - There is an NBME Q that asks about a patient's response to ADH who is on chronic NSAIDs, and the answer is "↔ Response to ADH" and "↔ urinary osmolality" (meaning no change). - 2CK Q gives patient with primary hyperparathyroidism and ↑↑ serum calcium + nephrogenic DI; Q asks cause of the DI → answer = hypercalcemia. High calcium can cause renal insensitivity to vasopressin. - Serum vs urinary values are the opposite of SIADH: - ↑ serum sodium (>145 mEq/L), ↑ serum osmolality, ↑ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - **PCT isotonic; JGA hypotonic; MCD hypotonic.** → The PCT is always isotonic no matter what; the JGA is always hypotonic no matter what; the MCD is clearly hypotonic in DI since we're not pulling free water out of the urine. - When we are trying to first diagnose DI, the first thing we do is fluid restriction, same as with SIADH. We want to see how serum/urinary values change first. - After we determine that the urine is staying dilute + the serum is staying concentrated, the next best step is desmopressin (analogue of vasopressin). If the urine gets more concentrated, (i.e., if the drug works), we know central DI is the diagnosis and we're merely deficient in ADH. - If desmopressin doesn't work, we know we have nephrogenic DI. I should point out that even in nephrogenic DI, desmopressin might work *but only very little,* whereas with central DI, administration will ↑↑ urinary osmolality robustly. It will be obvious on USMLE. But my point is, don't say, "Oh well desmopressin worked like 5% so we can't have nephrogenic DI here." - Treatment for central DI is therefore desmopressin. - Treatment for nephrogenic DI is NSAID + a thiazide. Sounds weird, but ↓ Na^+ reabsorption induced by thiazides in the early-DCT promote compensatory ↑ Na^+ reabsorption in the PCT, where water follows Na^+ and our net loss of fluid is less than without the thiazide. In healthy individuals, however, they *will* lose more net fluid with the thiazide. The NSAID presumably ↓ renal blood flow, which will ↓ GFR and ↓ net fluid loss. - There is difficult 2CK NBME Q where they say patient is on lithium + has ↑ urinary output, and fluid restriction is wrong answer to this question (I say hard because 9/10 times, fluid restriction is correct when it's listed); answer = NSAID + thiazide diuretic. The implication is, if it's obvious what the patient's diagnosis is already nephrogenic DI, going straight to Tx is acceptable.
PP	- Psychogenic polydipsia means the patient is simply drinking too much. - Both the urine and serum will be dilute. - Serum vs urinary values are the opposite of SIADH: - ↓ serum sodium (<135 mEq/L), ↓ serum osmolality, ↓ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - I'd say 3/4 Qs on USMLE are obvious and will say some psych patient is drinking lots to "clear himself from evil spirits," etc. - Probably 1/4 Qs won't be an obvious psych vignette, but will just show you the lab values where you have to say, "The urine and serum are *both* dilute, so this is psychogenic polydipsia." - First step in diagnosis is fluid restriction in order to see how urinary/serum values change.

Obscure miscellaneous conditions	
Bartter syndrome	- Disease that presents as though patient is on loop diuretic. - For whatever magical reason, is associated with juxtaglomerular cell hyperplasia.
Gitelman syndrome	- Disease that presents as though patient is on thiazide diuretic.

Hyper-terse summary of drugs causing HY renal conditions	
- Discussed all this stuff earlier. But students confuse this stuff a lot. So the summary is warranted.	
Acute tubular necrosis	- Aminoglycosides (gentamicin, tobramycin, amikacin). - Cisplatin.
Interstitial nephritis	- NSAIDs, β-lactams, cephalosporins, allopurinol.
Membranous glomerulonephropathy	- Dapsone, gold salts, sulfonamides.

Diuretics / Renal-related drugs	
	
Mannitol	- Osmotic diuretic that ↑ osmolality of urine, thereby retaining free water within the PCT and descending limb of loop of Henle. - Can be used as Tx for ↑ intracranial pressure after patient has been intubated + hyperventilated (↓ CO2 → ↓ cerebral perfusion). - Avoid in heart failure (transient ↑ in serum osmolality prior to renal excretion → ↑ retention of free water within vascular space → transient ↑ preload on heart). - USMLE asks about mannitol's effect on serum osmolality and ADH. This is a Q repeated twice on the NBMEs. They will show a graph same as below and ask what change will be expected with mannitol administration:  Answer = C; serum osmolality goes ↑, which then causes ADH to go ↑ (this is because ADH normally ↑ free water reabsorption in setting of ↑ serum osmolality).

Acetazolamide	- Carbonic anhydrase inhibitor (enzyme that enables $H_2CO_3 \leftrightarrow HCO_3^- + H^+$) that acts within the tubular cells of the PCT. - Classic use is altitude sickness → causes ↓ reabsorption of HCO_3^- in the PCT → results in ↑ urinary pH and volume + ↓ serum pH and volume. In altitude sickness, patients hyperventilate (↓ CO2 → ↑ pH) to take in more O2, but it takes a minimum of 12-24 hours for the kidney to ↓ HCO_3^- reabsorption, so acetazolamide forces the ↓ reabsorption by causing metabolic acidosis.
Expired tetracyclines	- Can cause Fanconi syndrome (↓ ability of PCT to reabsorb amino acids, glucose, bicarb, and phosphate).
Furosemide	- Classic loop diuretic that inhibits the apical $Na^+/K^+/2Cl^-$ ATPase symporter on the thick ascending limb. Apical means side of urine. Basolateral means side of blood. - Most efficacious diuretic at enabling fluid unloading for pulmonary and peripheral edema. Do not decrease mortality in cardiac patients. - Answer on USMLE for patient with heart failure who has ↓ O2 sats who "refuses to lie supine/back on the gurney" because he/she can't breathe. Q might say CXR or exam shows fluid 2/3 up the lung fields. - Application of loops can be confused by students with ACEi/ARB. For instance, the latter are used first-line to ↑ EF in heart failure even if O2 sats are low (↓ AT-II constrictive effect on arterioles → ↓ afterload → heart pumps easier), but loops are first-line if the vignette *specifically emphasizes* the dyspnea and low O2 sats of the patient as the focus. - HY point is that they can cause **hypokalemia.** What USMLE loves to do is tell you a patient is initiated on furosemide but it is insufficient + now needs a second diuretic → answer = anything that is potassium-sparing (i.e., ENaC inhibitor such as amiloride or triamterene; or aldosterone receptor antagonist such as spironolactone or eplerenone). - Can also cause ototoxicity (tinnitus, vertigo). - Loops ↑ urinary calcium. This is in contrast to thiazides, which ↓ it. - Ethacrynic acid is another loop you can be aware of. It is used in patients with sulfa allergies, since furosemide is a sulfa drug. It is also ototoxic.
Thiazides	- Hydrochlorothiazide (HCTZ) is HY example; chlorthalidone is "thiazide-like." - Inhibit the Na^+/Cl^- symporter on the apical membrane of the early-DCT. - ↑ calcium reabsorption in the DCT of the kidney. Used to ↓ recurrence of calcium stones by promoting ↑ reabsorption of urinary calcium. In turn, they can sometimes cause hypercalcemia. There is an NBME Q where they ask for MOA of HCTZ, where Na^+/Cl^- symporter inhibition isn't listed; answer is just "↑ calcium reabsorption." - Thiazides and dihydropyridine calcium channel blockers are used first-line for HTN in patients without any renal or cardiovascular issues. If patient has proteinuria, ↑ in creatinine or renin, or pre-diabetes or diabetes, ACEi or ARBs are used first. - Can be used in heart failure in patients with diabetic nephropathy who are already on ACEi and β-blocker. Sounds specific, but there's a new 2CK Q that has thiazide as correct over spironolactone in a diabetic. Spironolactone can cause hyperkalemia in patients who have worrisome kidney function. - Thiazides can cause gout (i.e., contraindicated in gout). - Offline NBME 20 wants you to know thiazides can cause galactorrhea (milky discharge from the nipples) via some obscure mechanism. Literature search shows it accounts for ~0.2% of adverse effects of thiazides, but count on NBME to ask it.
Amiloride, Triamterene	- ENaC inhibitors → block apical sodium channel in cortical collecting duct → ↓ Na^+ reabsorption → ↓ water reabsorption. - These are potassium-sparing, which means they do not ↓ serum K^+. This is because by inhibiting the apical ENaC channel, they indirectly inhibit the basolateral Na^+/K^+ ATPase → ↓ Na^+ reabsorption + ↓ K^+ secretion. - The answer on USMLE for a second diuretic given in a patient already on furosemide who needs additional fluid unloading. Be careful however. I've seen one Q where the patient is on furosemide, but the point of the Q is he/she needs

	HTN control, and the answer is a thiazide, not the ENaC inhibitor. You want to select an ENaC inhibitor specifically if the Q says, "We have a patient who has peripheral/pulmonary edema due to X cause + is already on furosemide; what do we do now?" → answer = amiloride or triamterene.
Spironolactone, Eplerenone	- Aldosterone receptor antagonists. - By blocking aldosterone receptor, they ↓ activity of the basolateral Na^+/K^+ ATPase → leads to indirect ↓ activity of apical ENaC → ↓ Na^+ reabsorption → ↓ water reabsorption. - Potassium-sparing. Same as with ENaC inhibitors, used for fluid unloading in patients who are already on furosemide in whom we worry about dropping their K^+ too much. - Used in heart failure up the hierarchy of meds – i.e., ACEi/ARB first-line, followed by adding β-blocker, followed by spironolactone (but if patient is diabetic with poor renal function, don't add spironolactone here; give thiazide instead as per new 2CK NBME). - Can be for aldosteronoma (Conn syndrome) prior to surgery. - Spironolactone can cause **gynecomastia** (anti-androgenic effect by blocking androgen receptors). - Eplerenone has ↓ risk of gynecomastia compared to spironolactone.
Lisinopril	- USMLE-favorite ACE inhibitor. - Prevents conversion of AT-I into AT-II in the lungs. - Used for HTN in patients with pre-diabetes, diabetes, atherosclerotic disease, or renal disease (I talk about this in HY Risk Factors PDF in more detail). - Can cause dry cough (ACE is aka bradykininase, so ACEi cause ↓ breakdown of bradykinin in lungs → cough). - Can ↑ serum K^+, since ↓ aldosterone synthesis. Aldosterone normally secretes K^+ in the distal kidney). - Avoid in hereditary angioedema.
Valsartan	- Angiotensin II receptor blocker (ARB). - Use-cases are identical on USMLE to ACEi (i.e., if you see both as answer choices to a question, they're usually both wrong because they're the "same"). - Doesn't cause dry cough the way ACEi do.
Tolvaptan, Conivaptan	- ADH receptor antagonists at V2 receptor in the medullary collecting duct. - Can be used for SIADH. - Remember, however, that demeclocycline is also answer on 2CK NBME for SIADH.

- 27M + dilated renal pelvis and collecting system + ureters not dilated; Dx? → USMLE wants "incomplete recanalization of proximal ureter."
- Four classic findings you see in nephrotic syndrome?
    - Proteinuria (usually high, i.e., >3g/24 hours)
    - Hypoalbuminemia (because of the proteinuria)
    - Peripheral edema (due to decreased intravascular oncotic pressure because of the hypoalbuminemia)
    - Hyperlipidemia (liver exports more apolipoproteins in an attempt to compensate for the low serum albumin)
- Four classic findings you see in nephritic syndrome?

  - **Hematuria** (due to glomerular inflammation allowing RBCs to escape into the urine)
  - Oliguria (<400mL/24 hours in adults; due to defective glomerular filtration with diminished GFR)
  - Azotemia (means high nitrogen compounds in the blood → blood urea nitrogen [BUN])
  - Hypertension (renal inflammation leads to microvascular disruption + upregulation of RAAS → high angiotensin-II probably more responsible for HTN than aldosterone, as AT-II is potent systemic vasoconstrictor, and patient need not have full-blown hyper-aldosteronism to have HTN in nephritic syndrome, in addition to HTN being able to occur in acute renal failure).

- Is there proteinuria in nephritic syndrome? → almost always yes, but just typically not as much as in nephrotic syndrome. The key differentiator between nephritic and nephrotic is the former has blood in the urine.
- Important nephrotic syndromes for USMLE? → membranous glomerulonephritis/nephropathy; minimal change disease (lipoid nephrosis); focal segmental glomerulosclerosis (FSGS); renal amyloidosis; diabetic glomerulonephropathy.
- Important nephritic syndromes for USMLE? → IgA nephropathy (Berger disease); post-streptococcal glomerulonephritis (PSGN);
- 12M + sore throat + blood in the urine 1-3 days later; Dx? → answer = IgA nephropathy, **not** post-streptococcal glomerulonephritis (PSGN).
- 12M + sore throat + blood in the urine 1-2 **weeks** later; Dx? → PSGN. This distinction between IgA nephropathy and PSGN is mega-HY.
- 12M + viral infection + red urine 2 days later; renal biopsy finding? → IgA deposition in the mesangium (IgA nephropathy).
- 12M + GI infection (diarrhea) + red urine 2 days later; Dx? → IgA nephropathy.
- 3F + violaceous lesions on her buttocks/thighs + arthralgias + abdominal pain + red urine; renal Dx? → IgA nephropathy. The tetrad of findings here is classic for Henoch-Schonlein purpura (HSP). The renal component is IgA nephropathy.
- "Wait I'm confused. How does HSP relate to IgA nephropathy?" → don't think of IgA nephropathy and HSP as distinct / unrelated conditions; they're on the same disease spectrum. While some patients might just get IgA nephropathy from a viral or GI infection, others might get the immune complex

deposits in the skin (purpura), immune complex deposits in joints (reactive arthritis; type III hypersensitivity), and mesenteric adenitis (abdo pain), leading to presentation of HSP. HSP is seen in pediatrics, whereas IgA nephropathy can easily occur in adults.

- 12F + crusty yellow lesions around the mouth for past week + today has red urine; Dx? → PSGN secondary to impetigo → give topical mupirocin to Tx the impetigo.
- Can you get PSGN from skin infections? → as per above, clearly → impetigo, cellulitis, and erysipelas, if caused by Group A Strep, can cause PSGN, yes. This is HY for the USMLEs. However it should be noted that rheumatic heart disease does **not** occur from cutaneous Strep infections; it occurs from Strep pharyngitis only.
- 28M + blurry vision and/or hearing issue + red urine; Dx? → Alport syndrome.
- Inheritance pattern of Alport? → X-linked recessive (according to retired NBME 19 for Step 1; I point this out because some sources say X-linked dominant, but the NBME says XR; the NBME always wins).
- Mechanism of Alport? → **mutation** in type IV collagen (in contrast to Goodpasture, which is **antibodies** against type IV collagen) → important in the lens of the eye + stereocilia within the cochlea + basement membranes within the kidney → classically "ear and/or eye problem + red urine" in a male. Question on retired NBME 19 for Step 1 gave the above 28M presentation, and the answer to the Q was "X-linked recessive."
- 13M + upper URTI + red urine after one week + decreased complement C3; Dx? → "acute glomerulonephritis" → yes, PSGN – same thing – but on one of the 2CK NBME's, they list the answer as just simply "acute glomerulonephritis," which for some reason throws people off.
- 13M + PSGN; renal biopsy shows what? → subepithelial deposits or "lumpy bumpy appearance."
- When is membranoproliferative glomerulonephritis (MPGN) the answer? → red urine in patient with hepatitis C or malignancy.
- "Splitting of glomerular basement membrane"; Dx? → MPGN. "Tram-track appearance."
- When is diffuse proliferative glomerulonephritis (DPGN) the answer? → red urine in patient with SLE.
- When is rapidly progressive glomerulonephritis (RPGN) the answer? → acute deterioration in renal function (high Cr, high BUN, oliguria) in someone who has a vasculitis (granulomatosis with polyangiitis [formerly Wegener], eosinophilic granulomatosis with polyangiitis [formerly Churg-Stauss], or microscopic polyangiitis) or Goodpasture syndrome.

- Biopsy shows what in RPGN? → fibrin crescents; parietal cell hyperplasia; leukocytic infiltrate → should be noted that the crescents on biopsy are extremely HY on Steps 1+2CK; on Step 1, knowing their composed of fibrin is high-yield.
- If the question is about Wegener, what will the renal biopsy often show? → "necrotizing" glomerulonephritis.
- Antibodies in Wegener vs CS vs MP? → Wegener = c-ANCA; CS + MP = p-ANCA. It should be noted that in real life, there is often overlap between the ANCAs, as well as times when patients may be ANCA-negative, but on the USMLE, the classic associations stand correct.
- Which antibodies are present in the ANCAs? → c-ANCA = anti-proteinase 3 (anti-PR3); p-ANCA = anti-myeloperoxidase (anti-MPO).
- Mechanism for Goodpasture? → antibodies against the alpha-3 chains of type IV collagen (anti-glomerular basement membrane antibodies; anti-GBM) → an NBME Q asks for mechanism, and answer is → "autoantibodies against host cell basement membranes."
- Diagnosis of Goodpasture? → biopsy shows linear immunofluorescence.
- How does Goodpasture present? → 40s male with hemoptysis + hematuria.
- How does Wegener present? → hematuria + hemoptysis + "head"-itis (i.e., any type of head finding, e.g., nasal septal perforation, mastoiditis, otitis, sinusitis).
- How does Churg-Strauss present? → asthma + eosinophilia; head findings rare.
- How does microscopic polyangiitis present? → basically just isolated hematuria in patient who's p-ANCA positive.
- Wrist- or foot-drop in someone with W, C-S, or MP; Dx? → Mononeuritis multiplex, which is palsy of one large nerve in multiple locations, such as the radial or peroneal, causing wrist or foot drop. Seen sometimes in vasculitides where you say, "Why is there neuropathy in this Wegener patient?" Etc.
- 38F + hematuria + hemoptysis + anti-PR3; Dx? → Wegener.
- 38F + hematuria + nasal septal perforation + c-ANCA; Dx? → Wegener.
- 38F + hematuria + p-ANCA; Dx? → MP.
- 38F + hematuria + anti-MPO; Dx? → MP.
- 38F + asthma + eosinophilia + nasal septal perforation; Dx? → Churg-Strauss (yes, "head"-itis is more rare, but the asthma + eosinophilia are overwhelmingly CS over Wegener; saw one question like this).

- 44M + hematuria + hemoptysis + no other findings; Dx? → Goodpasture most likely. If Wegener, they usually give you a "head-itis" finding, such as nasal septal perforation. If microscopic polyangiitis, usually just hematuria alone.
- 44M + hematuria + hemoptysis + renal biopsy shows necrotizing glomerulonephritis; Dx? → Wegener.
- 44M + hematuria + hemoptysis + linear immunofluorescence on renal biopsy; Dx? → Goodpasture.
- Alport syndrome vs Goodpasture key difference? → Alport = mutation in type IV collagen gene (*COL4A3/4/5*); Goodpasture = antibodies against type IV collagen.
- 12M + periorbital edema + red urine + low C3; Dx? → PSGN.
- 12M + periorbital edema + no blood in the urine + no other information; Dx? → minimal change disease (MCD; lipoid nephrosis).
- 12M + recent viral infection + periorbital/pedal edema + ascites; Dx? → MCD.
- 40M + Hodgkin lymphoma + renal condition (no blood in urine); Dx? → MCD → student says wtf? → MCD is almost always children following viral infection, but it's also seen in adults with Hodgkin. First time you hear this it's super-weird, but it's not a big deal.
- Tx for MCD? → steroids (highly effective)
- Sickle-cell + nephrotic syndrome; Dx? → FSGS.
- HIV / IV drug use + nephrotic syndrome; Dx? → FSGS.
- Interferon use + nephrotic syndrome; Dx? → FSGS.
- What does FSGS mean? → focal vs global; segmental vs diffuse; focal means a part of the kidney; diffuse means the whole kidney; segmental means part of a single nephron; global means an entire single nephron. So in FSGS, parts of some nephrons are involved. This makes sense for sickle-cell, for instance, where sickling will cause irregular microvascular pathology leading to non-uniform renal involvement.
- Tx for FSGS? → steroids (not as effective as compared to MCD).
- Sulfa drug + kidney issue + no blood in urine; Dx? → membranous glomerulonephropathy (MG).
- Biopsy finding in MG? → subepithelial deposits; "spike and dome" appearance.
- HepB or C + nephrotic syndrome; Dx? → MG → usually due to HepB.

- Autoantibodies in membranous glomerulonephropathy? → Sometimes patients are positive for anti-phospholipase A2 receptor antibodies.
- HepC + nephritic syndrome; Dx? → MPGN.
- Multiple myeloma + renal diagnosis? → renal amyloidosis → immunoglobulin light chains at high levels moving through the kidney (Bence Jones proteinuria) leads to deposition in the renal parenchyma. You should memorize the sentence: "Multiple myeloma is the most common cause of renal amyloidosis."
- First change in the kidney with diabetes? → hyperfiltration → increased filtered glucose pulls water with it. USMLE Q mentions guy with diabetes + polyuria; asks why he has increased urinary output → answer = "increased glomerular filtration rate."
- First *histologic* change in the kidney with diabetes? → thickening of the glomerular basement membrane → due to non-enzymatic glycosylation of basement membrane. If the question asks you for the first renal change seen overall in diabetes, select hyperfiltration.
- Amount of glucose reabsorbed in PCT? → 100% physiologically; glycosuria not seen until serum levels exceed around 180 mg/dL.
- Late finding seen histologically in diabetic nephropathy? → Kimmelstiel-Wilson nodules.
- What are KW nodules composed of? → hyaline → HY on Step 1 for some reason; don't confuse this fibrin, which is the answer for the crescents in RPGN.
- Example of type II diabetes drug that acts in the kidney? → Dapagliflozin (SGLT2 inhibitor in PCT; prevents reabsorption of glucose).
- What is hyaline arteriolosclerosis? → hyaline deposition in the arterioles of the kidney usually seen in diabetes → non-enzymatic glycosylation of vascular endothelium leads to leakage of plasma proteins into vessel walls.
- First drug given to diabetics with HTN or proteinuria? → ACEi (e.g., enalapril) or ARB (e.g., valsartan).
- Most common cause of chronic renal failure? → diabetes mellitus.
- Comment on which endogenous mediators affect diameter of the renal afferent vs efferent arterioles → prostaglandins dilate the afferent; angiotensin-II constricts the efferent.
- What do NSAIDs do to renal vasculature? → decrease dilation of the afferent arterioles (decreased prostaglandin synthesis doesn't cause constriction; we simply see attenuated degree of dilation).

- What do ACEi/ARBs do to renal vasculature? → decrease constriction of the efferent arterioles (decreased AT-II synthesis/binding doesn't cause dilation; we simply see attenuated degree of constriction). USMLE will show you pic of glomerulus + ask where NSAID or ACEi acts, so just simply be able to identify that.
- Student asks, "Why does AT-II act on the efferent arteriole? What's the physiologic reason for that?" → physiologically, AT-II increases in the setting of low blood volume → constriction of efferent arteriole → increased filtration fraction (GFR/RBF) → GFR can be maintained in the setting of reduced renal blood flow
- What do ACEi/ARB do to filtration fraction? → decrease it.
- Student asks, "If diabetes is most common cause of chronic renal failure, and renal failure is defined as low GFR, why would we give them ACEi or ARB then first-line if those drugs decrease filtration fraction?" → It's paradoxical, yes. Although you'd be decreasing filtration fraction with one of these drugs, you'd *also* be decreasing the rate at which glucose is being filtered across the basement membrane, thereby slowing the non-enzymatic glycosylation of the BM + progression of disease. Patients with end-stage renal disease are generally taken off ACEi/ARB, however early on these drugs are a mainstay of treatment.
- BUN/Cr ratio in pre-, intra-, vs post-renal failure? → >20 in pre-; <20 if not pre- → FA for Step 1 had stratified this out as <15 for intra- and 15-20 for post-, but I've seen at least three 2CK NBME/CMS Qs where the diagnosis was acute tubular necrosis and the BUN/Cr was in the 16s or 17s. So I've learned to just tell students: >20 = pre-; if <20, you simply know it's not pre-.
- Fractional excretion of sodium (FeNa) in pre-/intra-/post-renal failure? → need to know it's <1% in pre-renal; >1% in intra-/post-renal → the low FeNa in pre-renal is due to the PCT's attempt to reabsorb Na (water follows sodium) in low-volume state (or in FMD, RAS).
- Urine osmolality in pre-/intra-/post-renal failure? → concentrated (high; >500 mOsm) in pre-; dilute (low; <350 mOsm) in intra-/-post.
- When is pre-renal failure the answer? → Chronic heart failure, NSAID or diuretic use, dehydration from vomiting/diarrhea. Reduced perfusion over sub-acute to chronic time frame.

- Guy has MI + has low BP; NBME Q asks what is most likely to be seen (ask a bunch of reabsorption / secretion answers); answer = "increased potassium secretion" → low-volume status leads to RAAS upregulation and distal renal K wasting.
- When is intra-renal failure the answer? → ATN due to drugs (aminoglycosides, cisplatin), acute ischemia from acute blood loss (surgery, trauma), contrast nephropathy, rhabdomyolysis.
- Why does acute hypoxia cause acute tubular necrosis? → PCT is most susceptible to anoxic injury. Proximal convoluted tubules have high concentrations of Na-K-ATPase transporters with high oxygen demand.
- Lab animal is given 100% nitrogen in dumb experiment; most likely part of the kidney to experience anoxic injury (they list everything) → answer = "proximal tubule."
- Classic finding in urine with ATN? → muddy brown granular casts; it should be noted that general "granular casts" **are not specific to ATN.** There's an IM CMS Q with an elderly woman who has CVA tenderness + granular casts on U/A; answer is pyelo not ATN.
- USMLE Q gives guy who has MI requiring resuscitation + subsequent oliguria; then they ask what you see on microscopic examination of the kidney; answer = "degenerating epithelial cells and dirty brown granular casts" or "necrosis of epithelial cells in proximal convoluted tubules."
- Electrolyte disturbance in ATN? → first week is oliguric phase (hyperkalemia due to decreased filtration); weeks 2-3 are polyuric phase (hypokalemia due to increased kaliuresis from PCT cells not being able to reabsorb K yet) → btw, kaliuresis means urination of K (great word if you want to feel sophisticated).
- When is post-renal the answer? → classically BPH or distal obstruction secondary to strictures or malignancy.
- How does USMLE like to assess post-renal? → classically will give you BPH + show you a pic of hydronephrosis (massively dilated kidney), then they'll ask the most likely cause of this patient's condition; answer = "increased Bowman capsule hydrostatic pressure," or "increased tubular hydrostatic pressure." This answer is on several NBMEs.
- BPH Tx? → alpha-1 blocker (e.g., tamsulosin, terazosin) and/or 5-alpha-reductase inhibitor (i.e., finasteride); patients may receive mono- or dual therapy.

- Tx for prostatic adenocarcinoma? → flutamide + leuprolide administered together (if the USMLE forces you to choose a sequence, pick flutamide *then* leuprolide, but in practice they're administered together) → flutamide blocks androgen receptors; leuprolide is a GnRH receptor agonist (given continuously it shuts off LH + FSH secretion).
- When are hyaline casts the answer? → classically dehydration on USMLE.
- When are waxy casts the answer? → end-stage renal disease (not HY, but student occasionally asks).
- When are RBC casts the answer? → glomerulonephritis.
- When are WBC casts the answer? → pyelonephritis classically.
- Wtf is Tamm-Horsfall protein? → most common protein found in urine; not found in plasma; produced by tubular cells (classically thick ascending limb); no other real significance; more just the name pops up in Qbank rarely and student will ask about it.
- What is interstitial nephropathy? → aka interstitial nephritis, or tubulointerstitial nephritis, where there are WBCs (eosinophils) in the urine; patient may or may not present with a fine maculopapular rash → almost always due to beta-lactams, cephalosporins, NSAIDs, or allopurinol.
- 62M + 6 weeks nafcillin for MSSA endocarditis + WBCs on U/A; Dx? → interstitial nephropathy.
- 35F + uses ibuprofen daily for headaches + rash + eosinophils in urine; what do you see on biopsy? → NBME answer = "inflammatory infiltrates in the interstitium" = interstitial nephropathy.
- What is analgesic nephropathy? → renal damage due to NSAIDs → can be renal papillary necrosis, interstitial nephritis, or generalized edema from reduced renal blood flow. If renal papillary necrosis, will present with dark urine from ischemia. If interstitial nephritis, will present with eosinophils in urine as described previously (patient may not necessarily have a rash). Analgesic nephropathy is not a specific/different renal diagnosis. It is just an umbrella term that refers to any situation in which the kidney is fucked up due to NSAIDs.
- 80F + taking high doses of naproxen for 6 weeks for osteoarthritis + peripheral edema; mechanism of the edema? → Mechanism for edema in setting of NSAID use → USMLE answer = "increased renal retention of sodium," or "decreased renal excretion of sodium."
- Why would analgesic nephropathy cause increased renal retention of sodium? → NSAID knocks out COX → decreased prostaglandin synthesis → decreased diameter of afferent arterioles → decreased

renal blood flow → PCT attempts to increase water reabsorption to compensate for perceived low volume status → PCT accomplishes this by increasing Na reabsorption → water follows sodium → fluid retention + edema.

- Patient with analgesic nephropathy; USMLE Q asks mechanism in terms of NSAID's ability to inhibit constriction/dilation at afferent vs efferent arterioles; answer = "vasodilating prostaglandins at afferent arteriole," causing reduced renal blood flow.
- When is renal papillary necrosis the answer? → classically in sickle cell; can also from drugs like NSAIDs; urine will be dark → renal papillae in the medulla receive less blood flow, so small changes in the microvascular supply can lead to sloughing + necrosis of the medulla. You'll be able to contrast this disorder from diffuse cortical necrosis and acute tubular necrosis because the latter two conditions, in the context of ischemia, are associated with massive, acute events.
- First answer for Tx of all kidney stones on USMLE → hydration / fluids.
- Tx for calcium oxalate stones? → fluids, then alkalinize urine (citrate); prevent recurrence with thiazide (decrease urinary Ca).
- Tx for calcium phosphate stones? → fluids; prevent recurrence with thiazide (decrease urinary Ca). Calcium phosphate stones form at higher pH, so do not alkalinize; CaOxalate stones form at lower pH, which is why citrate helps.
- Tx for struvite (ammonium magnesium phosphate) stones? → fluids; acidify urine with ammonium chloride; surgery (because they're often huge).
- Tx for uric acid stones? → fluids + alkalinize urine with citrate; treat the underlying gout; avoid uricosurics such as probenecid indefinitely; avoid low-dose aspirin.
- Tx for cystine stones? → fluids; alkalinize urine with citrate.
- Who gets calcium oxalate stones? → classically those with intestinal malabsorptive disorders such as Crohn, Celiac, CF → malabsorption of fats means increased intraintestinal binding of fat to calcium → less calcium available to bind oxalate intraluminally → more oxalate is absorbed through intestinal wall. CaOxalate stones can also be seen in ethylene glycol (AntiFreeze) poisoning and hypervitaminosis C.
- Who gets struvite stones? → UTIs caused by urease (+) organisms. Proteus spp. is classic. Stones are coffin lid-shaped.

- Who gets uric acid stones? → generally uric acid over-producers such as in Lesch-Nyhan syndrome or tumor lysis syndrome; alcoholics (USMLE wants under-excretion as mechanism via which EtOH causes gout, but EtOH also can increase production); stones are radiolucent on non-contrast CT.
- Who get cystine stones? → patients with cystinuria caused by inability to absorb COLA (cysteine, ornithine, lysine, arginine) → two cysteines (-SH) joined via disulfide bond make one cystine (-S—S-).
- Any factoid to know about cystine stones? → AR disorder; generally young adults in USMLE vignette; nitroprusside urine test yields a magenta color; hexagon-shaped stones.
- What do I need to know about renal tubular acidosis type I (RTA I)? → decreased ability of the DCT to secrete protons (urine pH is >5.5) → leads to metabolic acidosis; patient does not have hyperkalemia. Can be associated with bilateral renal **parenchymal** stones (i.e., not within the lumina but instead in the actual meat of the kidney); can be seen in Sjogren syndrome.
- What do I need to know about renal tubular acidosis type II (RTA II)? → decreased ability of the PCT to reabsorb bicarb (urine pH is >5.5) → associated with carbonic anhydrase inhibitors and Fanconi syndrome (expired tetracyclines). Patient does not have hyperkalemia.
- What do I need to know about renal tubular acidosis type IV (RTA IV)? → either hyporeninemic hypoaldosteronism or aldosterone insensitivity at the kidney (urine pH <5.5) → patient will present with lab values resembling Addison disease, but the etiology will be chronic renal failure, not adrenal. For instance, 68F diabetic with biochemistry resembling Addison = RTA IV, not Addison → patient will have low serum sodium, high potassium, low bicarb (metabolic acidosis). Patient is hyperkalemic, in contrast to RTA I and II.
- USMLE Q asks “decreased absorption” vs “no change” for Fanconi syndrome with respect to glucose, amino acids, phosphate, bicarb → answer = decreased absorption for all four.
- Explain RAAS system. → reduced renal perfusion (low blood volume, renal artery stenosis, fibromuscular dysplasia) → juxtaglomerular cells secrete renin → renin cleaves angiotensinogen (produced by liver) in the plasma into angiotensin I → AT I goes to lung where it’s cleaved by ACE into AT II → AT II causes vasoconstriction in systemic arterioles (HTN) + at the efferent arterioles of the kidney (increases FF); it also goes to the zona glomerulosa in the adrenal cortex, where it upregulates aldosterone synthase, thereby converting 11-deoxycorticosterone into aldosterone.

- Where does aldosterone act in the kidney? → cortical collecting duct; binds to intracytosolic receptor, causing upregulation of basolateral Na/K-ATPase pumps (1, 2, 3 → for every 1 ATP utilized, 2 K secreted into tubular cell from blood; 3 Na reabsorbed from tubular cell into blood) → this causes decreased intra-tubular cell Na → creates a favorable high-low gradient from urine into the tubular cell → ENaC is upregulated secondarily on apical membrane to allow Na to move from urine into tubular cell.
- Another important point about renin secretion? → USMLE wants you to know that beta-1 activity causes renin release. Sometimes you'll get a Q where a patient has low blood volume and they'll ask which is true, and the answer is "increased renal sympathetic activity."
- Where does vasopressin (ADH) act in the kidney? → medullary collecting duct → inserts aquaporins on apical membrane → increased free water reabsorption.
- Urine + serum findings in SIADH → concentrated urine (high osmolality) + low serum sodium (<135 mEq/dL).
- Urine + serum findings in diabetes insipidus → dilute urine (low osmolality) + high serum sodium (>145 mEq/dL).
- Four causes of nephrogenic DI or USMLE? → lithium + demeclocycline are the two highest yield. NSAIDs and hypercalcemia can also cause it.
- Urine + serum findings in psychogenic polydipsia (patient drinks too much) → dilute urine (low osmolality) + low serum sodium (<135 mEq/dL).
- In DI, comment on the urine osmolality relative to plasma (i.e., hypotonic, isotonic, or hypertonic) for PCT, JGA, medullary collecting duct → PCT is always isotonic regardless of renal condition; JGA is always hypotonic regardless of renal condition (joins the early-DCT with the afferent arteriole); medullary collecting duct hypotonic. In SIADH, the latter would merely be hypertonic.
- What is the JGA? → juxtaglomerular apparatus contains JGC that secrete renin.
- What is the macula densa? → modified smooth muscle cells of the DCT that sense changes in serum sodium, thereby triggering the JGC to secrete renin; if volume status is low, PCT is attempting to reabsorb more water to compensate; in order to do so, it reabsorbs more Na, where water follows Na. This means that less sodium will still be left in the urine in the DCT. The macula densa will sense

that as a signal that the PCT is reabsorbing it because the patient is fluid deficient → triggers JGA to secrete renin.

- What do I need to know about mannitol? → osmotic diuretic that acts by retaining water within the nephron lumina; USMLE likes proximal straight tubule (short segment after the PCT) as the answer if it shows you a drawing and is very specific; if proximal straight tubule isn't shown or labeled, choose thin descending limb over PCT. Avoid in heart failure because administration transiently increases intravascular volume → increased preload going to RA → increased myocardial oxygen demand. Mannitol is used to treat increased intracranial pressure (pseudotumor cerebri) in some patients *after* intubation + hyperventilation have been instituted (decreased CO2 → decreased cerebral perfusion).
- What do I need to know about acetazolamide and the kidney? → carbonic anhydrase inhibitor that acts to prevent bicarb reabsorption → causes a transient metabolic acidosis that can be used to counteract the respiratory alkalosis secondary to altitude sickness (kidney cannot physiologically compensate fast enough so the drug forces the metabolic acidosis).
- What do I need to know about Loops? → Furosemide is standard first-line agent → does not decrease mortality in heart failure → used to fluid-unload in patients with generalized edema or symptomatic pulmonary edema → inhibits the Na/K/2Cl symporter on the apical membrane of the thick-ascending limb → causes hypokalemia notably, but is also associated with contraction alkalosis (RAAS kicks up and secretes protons distally when volume status is low because of the Loop); also causes increased urinary calcium and magnesium; classically ototoxic, especially when combined with aminoglycosides like gentamicin; furosemide is a sulfa drug, so patients with sulfa allergy can go on ethacrynic acid instead (not a low-yield drug; memorize that). Bumetanide is an alternative to furosemide in some patients. In general, be aware that loop diuretics are the most effective fluid-unloaders of all of the diuretics.
- How are calcium + magnesium reabsorbed in the thick ascending limb? → paracellular (between the cells).
- What is Bartter syndrome? → disease that presents as though the patient is indefinitely on loop diuretic (associated with JGC hyperplasia).
- What do I need to know about thiazides? → hydrochlorothiazide (HCTZ) is standard drug → inhibits the Na/Cl symporter on the apical membrane of the early-DCT → associated with decreased urinary

calcium and magnesium (in contrast to loops) → known to cause hyperGLUC (hyperglycemia, -lipidemia, -uricemia, -calcemia) → avoid in diabetics + patients with Hx of gout; can be used to prevent recurrent calcium nephro-/ureterolithiasis. Thiazides may be used to Tx HTN in patients who do not have cardiovascular disease or diabetes (in these patient groups, use ACEi or ARB initially instead). Chlorthalidone is an alternative to HCTZ.

- What is Gitelman syndrome? → disease that presents as though the patient is indefinitely on thiazide diuretic.
- What do I need to know about amiloride and triamterene? → simply that they're ENaC inhibitors and are potassium-sparing (meaning they don't cause kaliuresis).
- What do I need to know about spironolactone? → aldosterone receptor antagonist used in heart failure after the patient is already on ACEi (or ARB) and a beta-blocker. Can also be used in patients with Conn syndrome / primary hyperaldosteronism. USMLE loves asking spironolactone as the answer in patients who are already on furosemide who need another diuretic (furosemide fluid-unloads the best but causes potassium wasting, whereas spironolactone is potassium-sparing, so adding the latter on top of furosemide can limit potassium losses). Also very HY to know that spironolactone causes gynecomastia due to its anti-androgenic effects. Eplerenone is an alternative to spironolactone that carries much lower risk of gynecomastia. These drugs decrease mortality in heart failure (HY; in contrast to furosemide, which does not decrease mortality and is used for symptoms only).
- What do I need to know about tolvaptan/conivaptan? → vasopressin (ADH) receptor antagonists that can be used in SIADH (e.g., due to small cell lung cancer).
- What are the three main effects of parathyroid hormone (PTH) at the kidney?
    - Upregulates apical Ca^{2+} reabsorption in the late-DCT (increases Ca reabsorption).
    - Decreases phosphate reabsorption in the PCT.
    - Upregulates 1-alpha-hydroxylase in the PCT → increases conversion of inactive 25-OH-D3 to active 1,25-$(OH)_2$-D3 → active vitamin D3 then goes to small bowel to increase both Ca + PO4 absorption; active D3 also goes to bone to increase mineralization of osteoid into hydroxyapatite.

- What kind of PTH-related pathology is seen in renal disease? → secondary hyperparathyroidism is caused by chronic renal failure → kidney cannot reabsorb Ca in the DCT or synthesis 1,25-D3 in PCT (decreased intestinal absorption), leading to hypocalcemia. PCT goes up to compensate.
- Comment on vitamin D in relation to renal disease. → patient will have decreased 1,25-$(OH)_2$-D3 and increased 24,25-$(OH)_2$-D3 (the latter is an inactive storage form that is created by the shunting of 25-OH-D3; USMLE asks about 24,25 sometimes). Student then asks, "I don't get it though. If vitamin D deficiency is associated with decreased Ca + decreased PO4, and there's vitamin D deficiency in chronic renal failure, why are the electrolytes in CRF decreased Ca + increased PO4?" → answer is: the renal disease always wins; yes, there's decreased PO4 absorption in the small bowel, but there's also decreased filtration at the kidney, and the latter wins in terms of the arrows. That is:
    - Vitamin D deficiency due to renal failure: low Ca + high PO4.
    - Vitamin D deficiency not due to renal failure: low Ca + low PO4.
- What is renal osteodystrophy? → any bone problem caused by renal disease, but is usually synonymous with "renal-induced osteomalacia" due to renal effects on reducing active Vit D3.
- Patient with chronic renal disease has a pseudofracture of the hip; Dx? → osteomalacia → "pseudofracture" is pathognomonic for vitamin D deficiency; in this case, the pseudofracture is just a type of renal osteodystrophy that may be seen.
- Patient has renal failure + friction rub heard on auscultation of the chest; Dx + Tx? → uremic pericarditis (a type of fibrinous pericarditis) → USMLE Tx is immediate hemodialysis.
- What is the biochemical / acid-base disturbance seen in renal disease? → sodium is variable; potassium high; calcium low; phosphate high; bicarb low; pH low; CO2 low (compensation).
- What do I need to know about EPO and the kidney? → Kidney secretes it in response to anemia / low oxygen states (e.g., COPD). EPO deficiency occurs in renal failure, leading to anemia of chronic disease. EPO excess secretion can occur in renal cell carcinoma.
- Smoker + hypercalcemia + polycythemia + hematuria; Dx? → RCC (paraneoplastic EPO + PTHrp).
- Smoker + flank pain + red urine; Dx? → RCC.
- Most common type of RCC? → clear cell carcinoma (histo shows large clear cells on Step 1).

- Notable disease associated with RCC? → Von-Hippel Lindau (VHL) → RCC can be bilateral. VHL is AD, chromosome 3, and also associated with retinal + cerebellar hemangioblastomas; caused by constitutive activation of hypoxia-inducible factor.
- Renal cancer in children? → Wilms tumor.
- 3M + painless flank mass + no other info; Dx? → Wilms tumor.
- 6M + painless flank mass + seizure + MRI of brain shows periventricular nodules; Dx? → Tuberous sclerosis with renal angiomyolipoma.
- Anything else about Wilms tumor? → increased incidence in horseshoe kidney in Turner syndrome.
- Horseshoe kidney key point? → not only increased risk of Wilms tumor, but the kidney gets caught under the IMA.
- What is WAGR complex? → Wilms tumor, Aniridia, Genitourinary malformation, Retardation; caused by WT1 gene mutation.
- What is Denys-Drash syndrome? → Wilms tumor + male pseudohermaphroditism (secondary sex characteristics are female; primary sex characteristics are male); caused by WT1 gene mutation.
- What is Beckwith-Wiedemann syndrome? → Wilms tumor + neonatal hypoglycemia + macrosomia / macroglossia + hemihypertrophy +/- omphalocele; caused by WT2 gene mutation.
- Comment on AR vs AD polycystic kidney disease. → ARPKD is chromosome 6 and a pediatric condition; it's associated with hepatic fibrosis. ADPKD is chromosome 16 and presents in 30s or 40s; associated with HTN and circle of Willis saccular (berry) aneurysms.
- Most common extra-renal location for cysts in ADPKD? → liver.
- What's more important in management of ADPKD, serial BP checks or MRI angiogram circle of Willis? → answer = serial BP checks → renal cysts compress microvasculature, causing surges in RAAS.
- Comment on systemic sclerosis and the kidney? → systemic sclerosis (diffuse type) can cause renal crises with malignant HTN.
- 68M + diabetic + Hx of intermittent claudication + CABG + gradually increasing BP past two years; Dx? → renal artery stenosis.
- Cause of renal artery stenosis? → atherosclerosis. They will always give RAS in a patient who has cardiovascular disease.

- 23F + high aldosterone + high renin + high BP; Dx? → fibromuscular dysplasia → this is **not** renal artery stenosis; if you say RAS, that means atherosclerosis. FMD is a tunica media proliferative pathology resulting in RAS-like presentation; it's seen classically in young women, whereas RAS will be patients with CVD.
- How to Dx FMD? → MR angiography (HY on the USMLE) → NBME has actually said the buzz / colloquial phrase "string of beads" appearance seen on MR angiography.
- Patient with high BP is given an ACEi or ARB and gets a spike in either renin or creatinine; Dx? → FMD if young patient; RAS if patient with CVD. USMLE loves this point and it shows up on Steps 1 and 2CK. If you have FMD or RAS, renal blood flow is already reduced, so the kidney has a reduced capacity to remain within physiologic range for GFR → giving an ACEi or ARB will reduce filtration fraction and lead to a clinically significant reduction in GFR → renin + Cr go up.
- Hx of many pregnancies + downward movement of vesicourethral junction → stress incontinence.
- Tx of stress incontinence → pelvic floor exercises (Kegel); if insufficient → mid-urethral sling.
- Hyperactive detrusor or detrusor instability → urge incontinence.
- Need to run to bathroom when sticking key in a door → urge incontinence.
- Incontinence in multiple sclerosis patient or perimenopausal → urge incontinence.
- Tx of urge incontinence → oxybutynin (muscarinic cholinergic antagonist) or mirabegron (beta-3 agonist).
- Incontinence + high post-void volume (usually 3-400 in question; normal is <50 mL) → overflow incontinence.
- Incontinence in diabetes → overflow incontinence due to neurogenic bladder.
- Tx for overflow incontinence in diabetes → bethanechol (muscarinic cholinergic agonist).
- Incontinence in BPH → overflow incontinence due to outlet obstruction → eventual neurogenic bladder.
- Tx for overflow incontinence in BPH → insert catheter first; then give alpha-1 blocker of 5-alpha-reductase inhibitor; then TURP if necessary.
- Innervation of external urethral sphincter → pudendal nerve (somatic; voluntary).
- Innervation of internal urethral sphincter → hypogastric nerves (sympathetic).

- Innervation of detrusor muscle → pelvic splanchnic nerves (parasympathetic; S2-4).
- What is Brenner tumor? → ovarian tumor with bladder (transitional cell) epithelium.
- Exquisitely tender prostate on digital rectal exam → prostatitis.
- Prostatitis Tx → ciprofloxacin (fluoroquinolone → DNA gyrase / topoisomerase II inhibitor).
- Costovertebral angle tenderness + fever → pyelonephritis.
- Costovertebral angle tenderness + granular casts → pyelonephritis (correct, super-weird; NOT acute tubular necrosis; this is on 2CK NBME).
- Tx for pyelonephritis → ciprofloxacin, OR ampicillin + gentamicin.
- Is chronic pyelonephritis ever an answer? → usually young patient, i.e., 3-4-year-old, who has recurrent bouts of acute pyelo → need to know recurrent acute pyelo is what causes chronic pyelo. For Step 1 level, they will show an image of a small, scarred kidney and give you the above description, then the answer is simply "vesicoureteral reflux" as the mechanism. They might also ask what you see on biopsy; answer = "tubular atrophy." Ultrasound shows "broad scars with blunted calyces." The phrase "thyroidization of the kidney" is buzzy and more Qbank, not NBME. You need to walk away knowing that chronic pyelo will produce scarred renal calyces in someone who's had recurrent acute pyelo.
- Tx for simple UTI → nitrofurantoin is classic answer (need not be cystitis in pregnant women; this is listed as the correct answer in many 2CK-level NBME/CMS Qs); TMP/SMX is also classic combo.
- Q asks best initial Mx to prevent UTIs → answer = postcoital voiding. If unsuccessful, NBME wants "postcoital nitrofurantoin prophylaxis" next; if not listed, choose "daily TMP/SMX prophylaxis." The latter sounds incredibly wrong, and I agree, sounds outrageous, but it's correct on one of the obgyn CMS forms and everyone gets it wrong.
- 22F + Sx of dysuria for 6 months + anterior vaginal wall pain + U/A completely normal + afebrile; Dx? → chronic interstitial cystitis → answer on one of the obgyn forms. Must have at least 6 weeks of UTI-like Sx without any identifiable pathology.
- Tx for asymptomatic bacteriuria → if pregnant, must Tx. If not pregnant, do not Tx. Exceedingly HY for 2CK.

- Why must Tx asymptomatic bacteriuria in pregnancy? → progesterone slows ureteric peristalsis, increasing the risk of pyelo.
- 82F falls down stairs; urine dipstick shows 2+ blood and 3-4 RBCs/hpf on LM; Dx? → rhabdomyolysis → need to know that false (+) blood on urine dipstick is rhabdomyolysis (dipstick cannot differentiate myoglobinuria from actual RBCs).
- 22F has Sx of UTI + U/A completely normal; next best step? → test for chlamydia/gonorrhea (answer on one of the newer obgyn forms).
- 25M + mucopurulent urethral discharge + culture grows nothing; Dx? → chlamydia (obligate intracellular).
- Tx for chlamydia and gonorrhea on the Step → If chlamydia, give azithromycin or doxycycline. If gonorrhea, give ceftriaxone PLUS either azithro or doxy (must co-treat for chlamydia). If you don't see the gram (-) diplococci on light microscopy, you know gonorrhea likely isn't there, so you only need to treat for chlamydia. But if you see gonorrhea there, you don't know whether chlamydia is also there or not, so must treat for both.
- Most common type of bladder cancer? → transitional cell carcinoma.
- Most specific cause of bladder cancer? → industrial (aniline) dyes.
- Most common cause of bladder cancer? → smoking.
- Type of bladder cancer seen with Schistosoma haematobium infection → squamous cell carcinoma.
- What is Potter sequence? → renal agenesis leading to oligohydramnios → causes pulmonary hypoplasia in fetus.
- Most common genitourinary (GU) abnormality in neonatal males? → posterior urethral valves (PUV) → thin membrane(s) in the urethra preventing the outflow of urine.
- Male neonate + suprapubic mass + no passage of urine; Dx? → PUV.
- Neonatal male with PUV; next best step in Dx? → ultrasound then voiding cystourethrogram.
- 34M falls on balance beam + saddle injury + blood at urethral meatus; Dx + next best step in Mx? → urethral injury; do retrograde urethrogram.
- 16M + skiing accident + bruising and/or pain over a flank + red urine; Dx + next best step in Mx? → kidney injury; must do CT with contrast, **not** ultrasound. This is exceedingly HY on surgery forms and sounds wrong (i.e., "why would you do CT with contrast in someone with suspected renal injury if

contrast can cause nephrotoxicity?") → great question, but CMS/NBME/USMLE still want CT with contrast, so don't pick ultrasound.

- 16M + skiing accident + bruising and/or pain over a flank + no gross blood of urine (i.e, urine not red); next best step in Mx? → urinalysis to look for blood (super HY) → if urinalysis shows blood, do CT scan to check for renal injury. If U/A negative, renal injury unlikely, so choose "no further investigations indicated," **not** CT.
- 16M + skiing accident + bruising and/or pain over a flank + urinalysis is normal; next best step? → no further investigations indicated.
- USMLE Q gives you creatinine of 1.0 mg/dL in 65M, then tells you 10 years later his Cr is 1.2 mg/dL; Dx? → normal aging → creatinine clearance naturally decreases with age. There is no definitive cutoff for USMLE Qs, but the student should be aware that Cr of 1.5 or greater is always pathologic, and at Cr of 2.0, >90% of nephron function has already been lost. Normal range is 0.7-1.2, but with age can occasionally creep upwards of 1.4.
- USMLE Q says oophorectomy is being performed + ureter is damaged; during ligation of which vessel did the damage to the ureter occur? → answer = ovarian artery.
- What is water under the bridge? → ureter travels under the uterine artery in women and vas deferens in men.
- Kid + bloody diarrhea + petechiae + red urine; Dx? → hemolytic uremic syndrome (HUS) caused by EHEC O157:H7 or Shigella → HUS = triad of thrombocytopenia + schistocytosis (microangiopathic hemolytic anemia) + renal insufficiency.
- How does HUS contrast with TTP? → TTP is caused by a mutation that results in defective ADAMTS13, or antibodies against ADAMTS13, resulting in the inability to cleave vWF multimers → platelet clumping → similar progression as HUS. One of the points of contrast is that TTP is not toxin-induced, and TTP also tends to be a pentad of the HUS findings + fever + neurologic signs.
- What is hepatorenal syndrome? → liver failure causing renal failure; rare Dx but the correct answer USMLE wants is "normal renal biopsy."
- Endocarditis following a genitourinary procedure (e.g., TURP) or catheterization; organism? → USMLE wants enterococcus.



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